WIRSBO™

# A3030005
# DuoMix 201
# Installation Instructions

The WIRSBO DuoMix 201 provides control of two variable speed injection mixing devices in order to satisfy a two temperature system. The system may consist of a two temperature radiant floor heating application, or a combination of snow melting and radiant floor heating. The control is designed for use with radiant systems but can be easily integrated to other hydronic applications (baseboard, fan coil). The control is capable of providing boiler return protection in order to prevent boiler flue gas condensation.
The DuoMix 201 includes a large Liquid Crystal Display (LCD) which provides temperature read-out and adjustable settings.
For building space heating applications the control includes features such as Warm Weather Shut Down (WWSD), outdoor reset, pump exercising, and maximum supply temperature adjustment. In the snow melting mode the control protects the slab against thermal shock, has idling capabilities, Warm Weather Shut Down (WWSD), and Cold Weather Cut Off (CWCO).

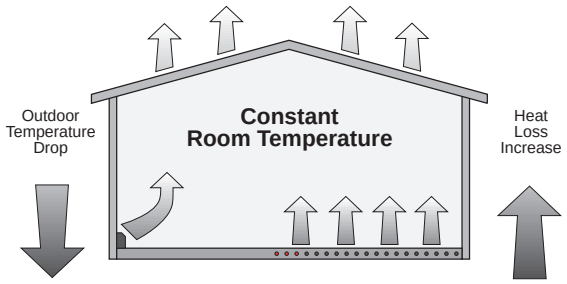
## Control Strategy

### OUTDOOR RESET

In order to maintain occupant comfort with a hot water heating system, the heat supplied to the building must equal the heat lost by the building.

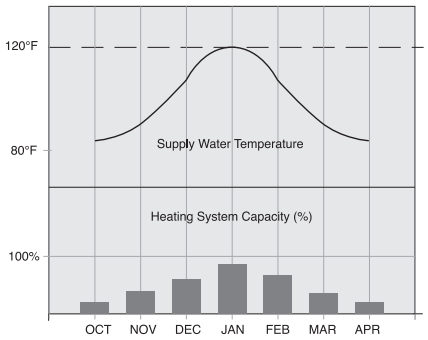
- The heat supplied to a building is proportional to the temperature of the water and the surface of the heating area. A small surface area such as baseboard radiators requires a higher water temperature than a larger surface area such as radiant floors.

- The heat supplied to a building/space can be changed by increasing or decreasing the supply water temperature. As the outdoor temperature falls the supply water temperature rises.




#### *Theory*

The calculated heat loss of a building or space is based on the coldest weather conditions the area experiences over the heating season. If the heat demand is analyzed as a percentage of the heating systems capacity, the heating system operates at or near maximum capacity for a small portion of the heating season. If the water temperature is controlled using a non-weather responsive tempering device, shown as the dotted line, the line is flat and does not flucuate with the demand for heat. Resetting the supply water temperature (solid line) allows for the supply water temperature to adjust itself to the demands on the heating system. Lowering the supply water temperature in warmer times of the heating season will allow the radiant zones to be open and run longer, providing increased comfort and efficiency.




#### *Water Temperature Optimization*

The Wirsbo DuoMix 201 uses actual data to program the control on start-up, eliminating the unknowns. Until now, resetting the supply water temperature to a radiant panel meant using terminology such as "curves" and "shifts". The DuoMix now eliminates manually selecting the proper curve based on the design data. From the user inputted data, the control creates a relationship (or ratio) between the outdoor temperature and the water temperature supplied to the radiant application. The control automatically adjusts the supply water temperature for every 1°F drop in outdoor temperature. In order to correctly reset the supply water temperature, a few parameters are required:

#### Outdoor Design Temperature

The Outdoor Design Temperature is used to calculate the heat load required at the most severe weather conditions expected during a normal heating season.

#### Design Supply Water Temperature

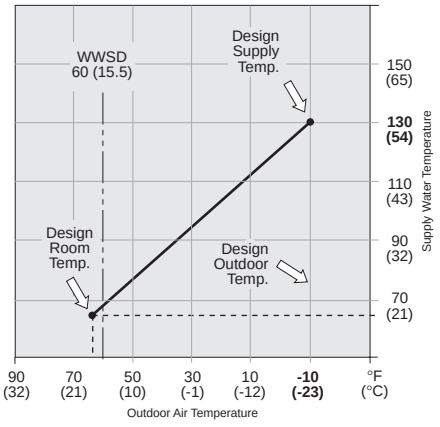
The Design Supply Water Temperature is the temperature of water supplied to the radiant panel(s) to meet the requirements of the heat load calculated at the outdoor design termperature.

#### Design Room Temperature

The Design Room Temperature is the indoor air temperature used in the heat loss calculations.

#### *Warm Weather Shut Down (WWSD)*

When the outdoor air temperature is warmer than the Warm Weather Shut Down setting, no additional heat is supplied to the building; therefore, the heating system can be shut down. The WWSD setting has no effect on the heating curve slope.




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### *Indoor Temperature Feedback*

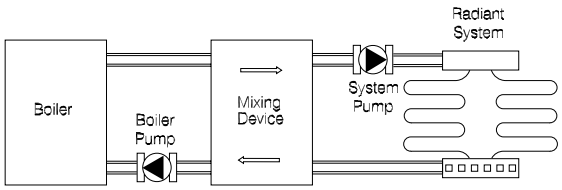
Most buildings have internal heat gains due to people, passive solar heating and mechanical or electrical equipment. If only the outdoor temperature is measured, the control cannot compensate for these internal heat gains and the building may overheat. In order to prevent overheating, indoor temperature feedback can be combined with the outdoor reset strategy. From this indoor temperature feedback, the control can match the supply water temperature to the heat loss of the building. If the indoor temperature is too warm, the control reduces the supply water temperature, and if the indoor temperature is too cold, the control increases the supply water temperature.

## SETPOINT OPERATION

Some loads, such as pool heating and floor warming, require a low fixed setpoint supply temperature. These loads should be regulated through a mixing device.

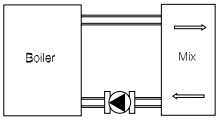
## MIXING OPERATION

The full range of water temperatures required through a heating season can be provided with a standard (non-condensing) boiler by incorporating a modulating mixing device into the system. Variable speed injection pumps can modulate both the system supply water temperature and the boiler return water temperature. The full modulation of water temperatures improves comfort in the building and also protects the boiler from cool return water.



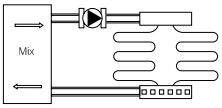

### *Boiler Protection*

Cool water is often returned to the boiler from snow melting or low temperature radiant floor heating systems. This cool boiler return water may cause the boiler to operate at such a low temperature that the flue gases condense. Alternatively, when the boiler surfaces are hot due to previous loads such as domestic hot water generation, the large temperature difference ($\Delta T$) between the boiler and its return water can cause the boiler to become thermally shocked. Proper protection of the boiler under these circumstances requires modulating mixing devices that can temporarily reduce the heating load. This is accomplished by reducing the speed of injection pumps.




### *System Maximum Supply*

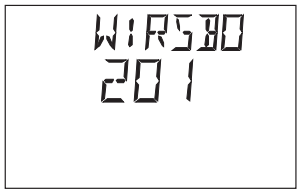
Hydronic radiant floor heating systems, usually operate at water temperatures that are below the minimum boiler return temperature. This is due to the large surface area of the floors which radiate a significant amount of heat at low water temperatures. Floor heating systems also have a maximum supply water temperature limit to protect the application in which it is installed. In such systems a mixing device is required to control and limit the supply water temperature to the radiant panel.




# Sequence of Operation

## POWERING UP THE CONTROL

When the WIRSBO DuoMix 201 is powered up, the words "WIRSBO 201" are displayed for 4 seconds. The control then displays the outdoor temperature and begins normal operation.

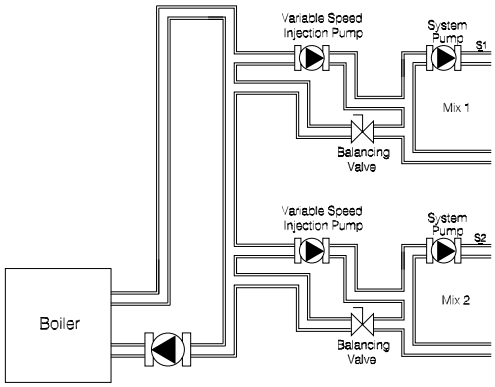



## WARM WEATHER SHUT DOWN (WWSD)

When the outdoor temperature rises above the WWSD 1 or WWSD 2 settings, the control shuts off the respective mixing system pump and variable speed injection pump. The *WWSD* element is displayed whenever both mixing loops are in WWSD. If no heat is required in any mixing system, the boiler enable relay is also turned off.

## MIXING OPERATION

The WIRSBO DuoMix 201 controls the supply water temperature to two different mixing loops (mixing 1 and mixing 2). The control provides the different water temperatures through two separate variable speed injection pump systems. The mixing 1 section can be chosen to operate in setpoint or outdoor reset mode. An optional indoor sensor may be installed in order to obtain indoor temperature feedback. The mixing 2 section can be chosen to operate in setpoint, outdoor reset, or snow melting mode.




### *Outdoor Reset Operation*

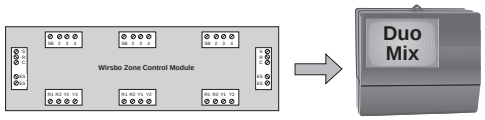
The DuoMix 201 can be used to provide outdoor reset to the mixing loops. The outdoor reset mode (*RST*) of operation on mixing 1 or mixing 2 must be selected through the display interface.



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### External Mixing Demand

When 24 or 120 V (ac) is applied to the *Mix1 Demand* terminals, the control turns on mixing pump 1 (*Mix1 Pmp*) and operates the variable speed injection pump 1 (*Var1 Pmp*). If 24 or 120 V (ac) is applied to the *Mix2 Demand* terminals, the control turns on mixing pump 2 (*Mix2 Pmp*) and operates the variable speed injection pump 2 (*Var2 Pmp*). The external demands may be obtained from the WIRSBO zone control, zone valve endswitches, or relay contacts.

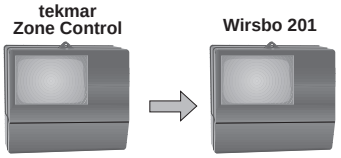



### Indoor Sensor

An indoor sensor may be connected to a mixing section in order to obtain indoor temperature feedback. The *SENSOR 1* or *2* setting must be set to *INDR* through the display interface. An external mixing demand is still required in order to provide heat.

### tekmar Zone Control

Outdoor reset can be combined with zoning. Through Room Temperature Units (RTU), a tekmar Zone Control can provide indoor temperature feedback to the WIRSBO 201 control. The 201 will then adjust the supply water temperature to satisfy the zone with the highest water temperature requirement. Heat to the remaining zones will be cycled on and off by the zone control using zone valves or pumps. If a tekmar Zone Control is connected to either the mixing 1 and/or mixing 2 sections, an external demand is not required, although it is still operational for additional zones that are controlled by thermostats.

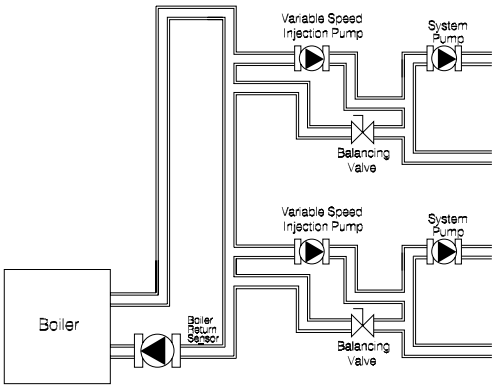



### *Boiler Enable*

When heat is required in the heating system the WIRSBO DuoMix 201 turns on the *Boiler Enable* relay. Once the *Boiler Enable* relay is turned on, a minimum of 3 minutes is allowed before turning the relay back off if the supply water temperatures are satisfied and the injection pumps have ramped down to 0%. If the heat demand is removed, the boiler is immediately turned off.

### *Boiler Protection*

A boiler sensor must be installed on the return line to the boiler in order to obtain boiler protection against flue gas condensation, if required. If the boiler return water temperature approaches the boiler minimum setting (*Boil RET*), the control maintains the *Boiler Enable* relay on and reduces the speed of the injection pump 2. If the boiler temperature still does not increase, then the speed of injection pump 1, will be decreased.

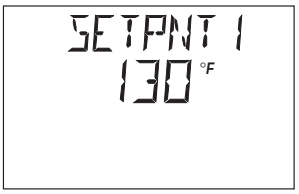



### *System Maximum Supply*

The WIRSBO DuoMix 201 has a maximum supply setting (*MAX SUP 1* and *MAX SUP 2*) for each mixing loop, which sets an upper operating limit to the mixed supply water temperatures for the application in which the tubing is installed. This value is different than the Design Supply Setting (*DSN SUP1* and *DSN SUP2*).

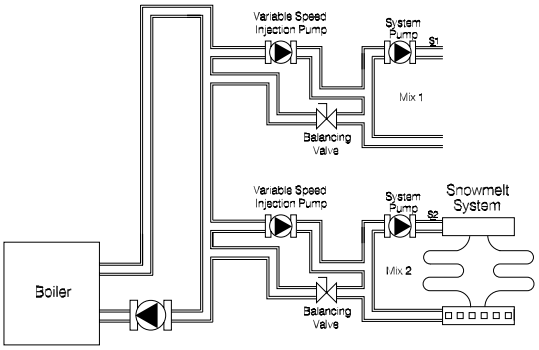
## SETPOINT OPERATION

The WIRSBO DuoMix 201 can be used to provide one or two fixed water temperatures to the mixing loops. The setpoint (*SETP*) mode of operation on mixing 1 or mixing 2 must be selected through the display interface. When 24 or 120 V (ac) is applied to the *Mix1 Demand* terminals, the control turns on the mixing pump 1 (*Mix1 Pmp*) and operates the variable speed injection pump 1 (*Var1 Pmp*) in order to supply the adjusted *SETPNT 1* water temperature. The mixing 2 section operates in the same manner.




## SNOW MELTING OPERATION

The mixing 2 section on the DuoMix 201 may be used to control a snow melting system. The snow melting mode of operation must be selected through the display interface (*SNOW*). When the *Start* button is pressed and held for 1 second on the control or on the Remote Module Unit 040, the control enters the snow melting mode of operation and displays the *RUN TIME*. The *RUN TIME* can now be adjusted for the desired operating time. The snow melting can also be enabled by the Remote Snowmelt Enable Kit A3040039. At any time during the snow melting session, the melting time remaining may be viewed. (See View Menu). An external *Mix 2 Demand* for 4 seconds is equivalent to pressing the *Start* button. If the Snowmelt Enable Kit is not desired, a Slab Sensor A3060072 and a Universal Sensor A3060071 are required for snow melt operation.

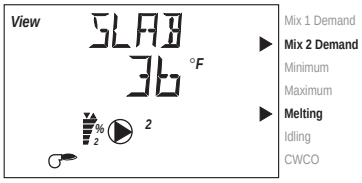





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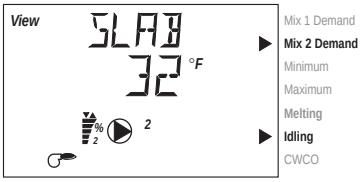
### *Melting mode*

Once in melting mode, the DuoMix 201 maintains the slab at the programmed *MELTING* temperature by controlling the speed of the injection pump 2 (*Var2 Pmp*). Whenever the control is in the melting mode, the mixing pump (*Mix2 Pmp*) is turned on and the *Melting* pointer is displayed.




### *Idling mode*

When the melting system starts off from a cold temperature, the time required for the slab to reach the melting temperature can be excessive. To decrease this start up time, the slab can be maintained at an idling temperature until melting is required. The idling feature is also useful for preventing frost and light ice formation. When the control is in idling mode, the slab is maintained at the IDLING temperature and the *Idling* pointer is turned on.

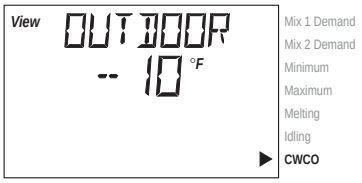



### *Warm Weather Cut Off (WWCO)*

If the slab and outdoor temperatures rise above the *MELTING* temperature, heating the slab is no longer required. The control therefore shuts down the melting system. When the outdoor or slab temperature drops below the *MELTING* temperature, the control exits *WWCO* and continues with normal operation.

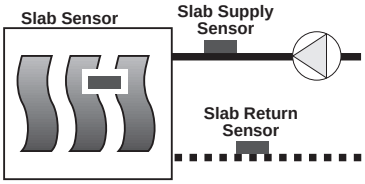
### *Cold Weather Cut Off (CWCO)*

Maintaining the slab at a *MELTING* or *IDLING* temperature in extremely cold weather can be expensive and may even be impossible. The control therefore turns the melting system off when the outdoor temperature drops below the *CWCO* setting and turns on the *CWCO* pointer. If the slab is at the desired *MELTING* temperature, the control will not enter the *CWCO* mode. If the an enable signal is given during the *CWCO* mode, the control continues to melt once out of the *CWCO* mode.



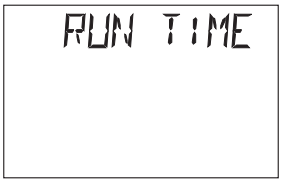

### *Slab Protection*

To protect the slab from cracking due to thermal stresses, the control limits the rate of heat applied to the slab through a $\Delta T$ *MAX* setting. The $\Delta$T represents the difference between the measured slab supply and return fluid temperatures. If this temperature difference approaches the $\Delta T$ *MAX* setting, the control operates the variable speed injection pump 2 to maintain the $\Delta$T at the $\Delta T$ *MAX* setting. The *Maximum* pointer is turned on when the target $\Delta$T is greater than the $\Delta T$ *MAX* setting.




#### Ramping the $\Delta$T during Melting System Start Up

When the control is in melting mode, the control maintains the maximum $\Delta$T and slowly increases the slab temperature to prevent thermal shock of the slab.




### *Run Time*

Prior to the control entering the snow melting mode, a *RUN TIME* should be selected. The snow melting time will start counting down once the slab is up to melting temperature. If the *Start* button on the control or the Remote Display 040 is pressed while snow melting is enabled and the slab is up to temperature, the run time counter is reset. Pressing the stop button on either control will end the snow melting operation. Using the Snow Melting Enable Kit A3040039, pressing the button during warm up or run time will end the snow melt operation.

### EXERCISING OF THE PUMPS

To prevent the pumps from seizing after long inactive periods, all pumps are turned on after every 3 days of no operation. The *Mix1 Pmp* and *Mix2 Pmp* pump relays are turned on for 20 seconds and the variable speed pump outputs are ramped up to 100% speed. During exercising, the LCD screen flashes "EXERCISE" and the *Test* LED flashes. Exercising can be stopped by pressing the *Test* button.



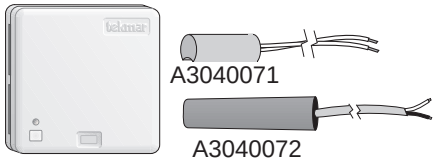
A3040040

### REMOTE DISPLAY A3040040 (tN2). AVAILABLE OCTOBER 1998

A Remote Display A3040040 can be used to duplicate the control display in a remote location. All settings can be remotely adjusted.

### SNOW MELT ENABLE KIT A3040039 (tN2)

The Snow Melt Enable Kit is used for snow melting applications. The kit includes a Remote Start/Stop Module, Mixing 2 Return Sensor A3040071, and a Slab Sensor A3040072. When a Remote Start/Stop Module is connected to the 201, the LED (Light Emitting Diode) flashes green or red. When the button is pressed, the control enters the snow melting mode. During slab warm up, the LED flashes green. Once the slab is up to temperature, the LED remains green continuously. If an error occurs in the control, the LED flashes red.




## Installation



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***Caution***

***Improper installation and operation of this control could result in damage to the equipment and possibly even personal injury. It is your responsibility to ensure that this control is safely installed according to all applicable codes and standards. This electronic control is not intended for use as a primary limit control. Other controls that are intended and certified as safety limits must be placed into the control circuit.***

## STEP ONE — GETTING READY

Check the contents of this package. If any of the contents listed are missing or damaged, please contact your wholesaler or Wirsbo sales representative for assistance.

*Type 201 includes:*
- One Wirsbo DuoMix 201 (A3030005)
- Three Universal Sensors (A3060071)
- One Outdoor Sensor (A3060070)

***Note*** *Carefully read the Sequence of Operation sections in all applicable brochures to ensure that you have chosen the proper control for your application.*

## STEP TWO — MOUNTING THE CONTROL AND SENSORS

### *Installing the Control*

- Remove the control from its base by pressing down on the release clip in the wiring chamber and sliding the control upwards.
- Mount the base directly onto the wall, DIN rail or an electrical box.

### *Installing the Outdoor Sensor A3060070*

- Remove the screw and pull the front cover off the sensor enclosure.

- The A3060070 can either be mounted directly onto a wall or a 2" x 4" electrical box. When the A3060070 is wall mounted, the wiring should enter through the back or bottom of the enclosure. Do not mount the A3060070 with the conduit knockout facing upwards as rain could enter the enclosure and damage the sensor.

- The A3060070 should be mounted on a wall which best represents the heat load on the building (a northern wall for most buildings and a southern facing wall for buildings with large south facing glass areas). The A3060070 should not be exposed to heat sources such as ventilation or window openings.

- In order to prevent heat that is transmitted through the wall from affecting the sensor reading, it may be necessary to install an insulating barrier behind the enclosure.

- The A3060070 should be installed at an elevation above the ground that will prevent accidental damage or tampering.

### *Installing the Universal Sensor A3060071*

- The sensor can be strapped directly to the pipe using the cable tie provided. Insulation should be placed around the sensor to reduce the effect of air currents on the sensor measurement.

- The Universal Sensor A3060071 should be placed downstream of a pump or after an elbow or similar fitting. This is especially important if large diameter pipes are used as the thermal stratification within the pipe can result in erroneous sensor readings. Proper sensor location requires that the fluid is thoroughly mixed within the pipe before it reaches the sensor.

## STEP THREE — ROUGH-IN WIRING

All electrical wiring terminates in the control base wiring chamber. The base has standard 7/8" (22 mm) knockouts which accept common wiring hardware and conduit fittings. Before removing the knockouts, check the wiring diagram and plan the routing of wires. The control base can be installed on a 4-3/4” x 4-3/4” electrical (dryer or range) box to simplify line voltage wiring.

**Power must not be applied to any of the wires during the rough-in wiring stage.**

- Install the Outdoor Sensor , Boiler Return Sensor, and the two Supply Sensors and run the wiring back to the control.

- If a 10K Indoor Sensor is used, install the Indoor Sensor(s) according to the installation instructions supplied with the sensor and run the wiring back to the control.

- If a tekmar Zone Control is used, run two wires from the Zone Control to the 201.

- If a Snow Melt Enable Kit A3040039 or a Remote Display Module 040 is used, run two wires from the unit to the 201.

- Run wiring from the other system components (pumps, mix demands, boiler, etc.) to the control. Multi-strand 16 AWG wire is recommended for all 120 V (ac) wiring due to its superior flexibility and ease of installation into the terminals.



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• Run wires from the 120 V (ac) power to the control. *Use a clean power source to ensure proper operation.*

## STEP FOUR — ELECTRICAL CONNECTIONS TO THE CONTROL

The installer should test to confirm that no voltage is present at any of the wires. Push the control into the base and slide it down until it snaps in firmly.

### *Powered Input Connections*

#### 120 V (ac) Power (*Power N & L* )

Connect the 120 V (ac) power supply to terminals *Power N — L* (15 and 16).

#### Mixing Demand (*Mix1 Demand & Mix2 Demand*)

In order for the control to provide heat, a powered external demand must be applied to the demand terminals (except when a tekmar Zone Control is used). Connect the 24, 120, or 240 V (ac) demand circuit to the *Mix1 Demand* terminals to operate the mixing loop 1. Connect the 24, 120, or 240 V (ac) demand circuit to the *Mix2 Demand* terminals to operate the mixing loop 2. See wiring on pages xx, xx.



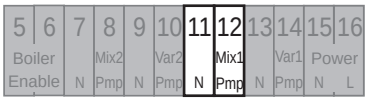




### *Output Connections*

#### Mixing 1 System Pump (*Mix1 Pmp*)

Connect one wire from the mixing 1 system pump to the *Mix1 Pmp* (12) terminal on the control. Connect the second wire on the mixing 1 system pump to the *N* (11) terminal. The control provides 120 V (ac) to the *Mix1 Pmp* contact whenever heat is required.




#### Mixing 2 System Pump (*Mix2 Pmp*)

Connect one wire from the mixing 2 system pump to the *Mix2 Pmp* (8) terminal on the control. Connect the second wire on the mixing 2 system pump to the *N* (7) terminal. The control provides 120 V (ac) to the *Mix2 Pmp* contact whenever heat is required.



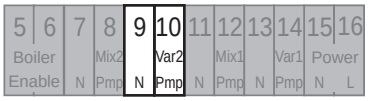

#### Variable Speed Injection Pumps (*Var1 Pmp & Var2 Pmp*)

The 201 can vary the speed of two permanent capacitor, impedance protected or equivalent pump motors that have a locked rotor current of less than 2.4 A. Most small wet rotor circulators are suitable as described in Essay E 021. *The variable speed output must not be used on pumps which have a centrifugal switch.* The 201 has two internal overload protection fuses which are each rated at 2.5 A 250 V (ac). These fuses are not field replaceable. Contact WIRSBO if a fuse is blown.
**Note:** If the combined current load of the variable speed injection pumps is 2.6 A, the ambient temperature should not exceed 122°F (50°C). If the combined load is 4.3 A, then the ambient temperature should not exceed 104°F (40°C).

Connect one wire from the variable speed injection pump 1 to the *Var1 Pmp* (14) terminal on the control. The other wire on the variable speed injection pump must be connected to the *N (*Neutral) (13) terminal. The control varies the power to the pump in order to change its speed.




Connect one wire from the variable speed injection pump 2 to the *Var2 Pmp* (10) terminal on the control. The other wire on the variable speed injection pump must be connected to the *N (*Neutral) (9) terminal.




#### Boiler Enable

Connect the 120 V (ac) or 24 V (ac) Boiler circuit to terminals *Boiler Enable* (5 and 6). The 201 closes a dry relay contact between these terminals when boiler operation is required.

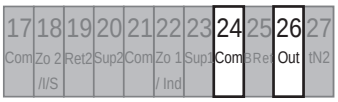



### *Sensor and Unpowered Input Connections*

***Do not apply power to these terminals as this will damage the control.***

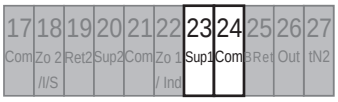
#### Outdoor Sensor (*Out*)

Connect the two wires from the Outdoor Sensor 070 to the terminals *Com — Out* (24 and 26). The Outdoor Sensor measures the outdoor air temperature.



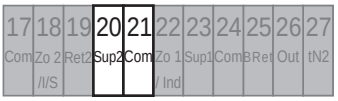

#### Supply 1 Sensor (*Sup1*)

Connect the two wires from the mixing 1 Supply Sensor 071 to terminals *Com — Sup1* (23 and 24). The Supply 1 Sensor measures the mixed supply water temperature in the mixing 1 section.




#### Supply 2 Sensor (*Sup2*)

Connect the two wires from the mixing 2 Supply Sensor 071 to terminals *Sup2 — Com* (20 and 21). The Supply 2 Sensor measures the mixed supply water temperature in the mixing 2 section.

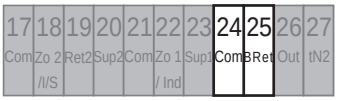





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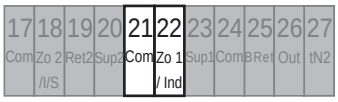
### Boiler Return Sensor (*BRet*)

Connect the two wires from the Boiler Return Sensor 071 to terminals *Com — BRet* (24 and 25). The Boiler Return Sensor measures the boiler return water temperature.

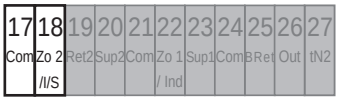



### Zone Control Input (*Zo 1 & Zo 2*)

If a tekmar Zone Control is used on the mixing 1 section, connect the wire from the *Com* terminal on the Zone Control to terminal *Com* (21) on the 201. Connect the wire from the *Zo Out* terminal on the Zone Control to terminal *Zo 1 / Ind* (22) on the 201.




If a tekmar Zone Control is used on the mixing 2 section, connect the wire from the *Com* terminal on the Zone Control to terminal *Com* (17) on the 201. Connect the wire from the *Zo Out* terminal on the Zone Control to terminal *Zo 2 / I/S* (18) on the 201.

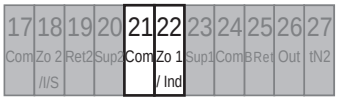


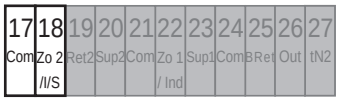

**Note** The wires from the Zone Control are polarity sensitive. The system will not operate if the wires are reversed.

### Indoor Sensors (Zo 1 / Ind & Zo 2 /I/S)

Indoor Sensors provide indoor temperature feedback to the control.

If an indoor sensor is used on the mixing 1 section, connect the two wires from the Indoor Sensor to the *Com – Zo 1/ Ind* (21 and 22) terminals on the control.
If an indoor sensor is used on the mixing 2 loop, connect the two wires from the Indoor Sensor to the *Com – Zo 12/ I/S* (17 and 18) terminals on the control.




### Slab Sensor (Zo 2 /I/S)

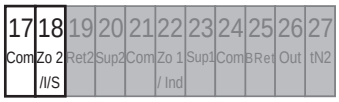
Slab Sensor measures the slab temperature.

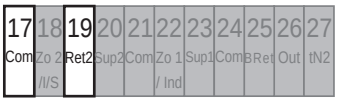
If the mixing 2 section is used for snowmelting, connect the two wires from the Slab Sensor to the *Com – Zo 2/ I/S* (17 and 18) terminals on the control.

### Slab Return Sensor 2 (Ret2)

Slab Return Sensor measures the return glycol temperature from the slab.

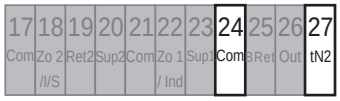
If the mixing 2 section is used for snowmelting, connect the two wires from the Slab Return Sensor to the *Com – Ret2* (17 and 19) terminals on the control.



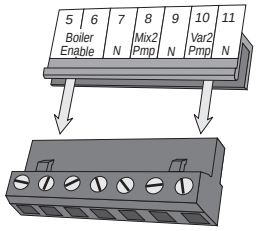

### tN2 Connection (Snow Melt Enable Kit A3040039 Remote Display Module 040)

Connect the first wire from the *Com* (24) terminal on the 201 to the *Com* terminal on the 040 or the Snow Melt Enable unit. Connect the second wire from the *tN2* (27) terminal on the control to the *tN2* terminal on the 040 or the Snow Melt Enable Module.




## STEP FIVE — TESTING THE WIRING

***Each terminal block must be unplugged from its header on the control before power is applied for testing. Pull straight down to unplug the terminal block.***

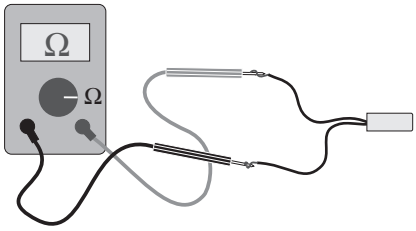



The following tests are to be performed using standard testing practices and procedures and should only be carried out by properly trained and experienced persons.

A good quality electrical test meter, capable of reading from at least 0 — 300 V (ac) and at least 0 — 2,000,000 Ohms, is essential to properly test the wiring and sensors.

### *Test the Sensors*

The actual temperature must be measured with either a good quality digital thermometer, or if a thermometer is not available, a second sensor can be placed alongside the one to be tested and the readings compared.




First measure the temperature using the thermometer and then measure the resistance of the sensor at the control. The wires from the sensor must not be connected to the control while the test is performed. Using the chart below, estimate the temperature measured by the sensor. The sensor and thermometer readings should be close. If the test meter reads a very high resistance, there may be a broken wire, a poor wiring connection or a defective sensor. If the resistance is very low, the wiring may be shorted, there may be moisture in the sensor or the sensor may be defective. To test for a defective sensor, measure the resistance directly at the sensor location with the wires disconnected.

Do not apply voltage to a sensor at any time as damage to the sensor may result.



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| Temperature | | Resistance | Temperature | | Resistance | Temperature | | Resistance | Temperature | | Resistance |
|---|---|---|---|---|---|---|---|---|---|---|---|
| °F | °C | Ω | °F | °C | Ω | °F | °C | Ω | °F | °C | Ω |
| -50 | -46 | 506,000 | 20 | -7 | 47,100 | 90 | 32 | 7,400 | 160 | 71 | 1,700 |
| -40 | -40 | 337,000 | 30 | -1 | 34,400 | 100 | 38 | 5,780 | 170 | 77 | 1,390 |
| -30 | -34 | 228,000 | 40 | 4 | 26,700 | 110 | 43 | 4,730 | 180 | 82 | 1,180 |
| -20 | -29 | 166,000 | 50 | 10 | 19,900 | 120 | 49 | 3,740 | 190 | 88 | 977 |
| -10 | -23 | 116,000 | 60 | 16 | 15,000 | 130 | 54 | 3,100 | 200 | 93 | 838 |
| 0 | -18 | 86,500 | 70 | 21 | 11,900 | 140 | 60 | 2,490 | 210 | 99 | 701 |
| 10 | -12 | 61,700 | 80 | 27 | 9,170 | 150 | 66 | 2,010 | 220 | 104 | 606 |

## *Test the Power Supply*

Make sure exposed wires terminals are not in contact with other wires or grounded surfaces. Turn on the power and measure the voltage between the *Power N — L* (15 and 16) terminals using an AC voltmeter, the reading should be between 110 and 130 V (ac).

## *Test the Powered Inputs*

### Mixing Demand (*Mix1 Demand & Mix2 Demand*)

If a mixing demand is used on the mixing 1 section, measure the voltage between the *Mix1 Demand* (1 and 2) terminals. When a call for heat on the mixing 1 section is present, you should measure between 24 and 240 V (ac) at the terminals. When the demand is gone, you should measure less than 5 V (ac).

If a mixing demand is used on mixing 2 section, measure the voltage between the *Mix2 Demand* (3 and 4) terminals. When a call for heat on the mixing 2 section is present, you should measure between 24 and 240 V (ac) at the terminals. When the demand is gone, you should measure less than 5 V (ac).

## *Test the Outputs*

### Mixing 1 System Pump (*Mix1 Pmp*)

If a mixing system pump is used in the mixing 1 section, make sure power to the terminal block is off and install a jumper between the *N – Mix1 Pmp* (11 and 12) terminals. When power is applied to these terminals, the mixing system pump *Mix1 Pmp* should start. If the pump does not turn on, check the wiring between the terminal block and the pump and refer to any installation or troubleshooting information supplied with the pump. If the pump operates properly, disconnect the power and remove the jumper.

### Mixing 2 System Pump (*Mix2 Pmp*)

If a mixing system pump is used in the mixing 2 section, make sure power to the terminal block is off and install a jumper between the *N – Mix2 Pmp* ( 7 and 8) terminals. When power is applied to these terminals, the mixing system pump *Mix2 Pmp* should start. If the pump does not turn on, check the wiring between the terminal block and the pump and refer to any installation or troubleshooting information supplied with the pump. If the pump operates properly, disconnect the power and remove the jumper.

### Variable Speed Injection Pumps (*Var1 Pmp & Var2 Pmp*)

If the variable speed injection pump of the mixing 1 section is used, make sure power to the terminal block is off and install a jumper between the terminals *N – Var1 Pmp* (13 and 14). When power is applied to terminals, the variable speed injection pump should operate at full speed. If the pump does not operate, check the wiring between the terminal block and the pump and refer to any installation or troubleshooting information supplied with the pump. If the pump operates properly, disconnect the power and remove the jumper.

If the variable speed injection pump of the mixing 2 section is used, make sure power to the terminal block is off and install a jumper between the terminals *N – Var2 Pmp* (9 and 10). When power is applied to terminals, the variable speed injection pump should operate at full speed. If the pump does not operate, check the wiring between the terminal block and the pump and refer to any installation or troubleshooting information supplied with the pump. If the pump operates properly, disconnect the power and remove the jumper.

### Boiler Enable

If a boiler is connected to the *Boiler Enable* (5 and 6) terminals, make sure power to the boiler circuit is off and install a jumper between the terminals. When the boiler circuit is powered up, the boiler should fire. If the boiler does not turn on, refer to any installation or troubleshooting information supplied with the boiler and/or Boiler Control. If the boiler operates properly, disconnect the power and remove the jumper.

## *Connect the Control*

• Make sure all power to the devices and terminal blocks is off and remove any remaining jumpers from the terminals.

• Reconnect the terminal blocks to the control by carefully aligning them with their respective headers on the control and then pushing the terminal blocks into the headers. The terminal blocks should snap firmly into place.



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• If required, install the supplied safety dividers between the unpowered sensor inputs and the powered 120 V (ac) or 24 V (ac) wiring chambers.

**• Do not apply power to the control until the DIP switches are properly set for your application. See the Settings section of this brochure for details on how to set the DIP switches.**

• Once the settings are complete, apply power to the control. The operation of the control on power up is described in the Sequence of Operation section of this brochure.

## Settings

Before adjusting the control settings, read through the Sequence of Operation section of this brochure to ensure that you understand how the control operates.

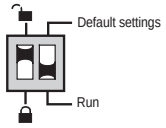
### STEP SIX — ESSENTIAL CONTROL SETTINGS

#### *Lock- Unlock DIP Switch*

The Lock-Unlock DIP switch is used to prevent unauthorized persons from tampering with the settings. When the DIP switch is set to Lock ( 🔒 ) none of the settings may be adjusted. To adjust a setting simply switch the DIP switch to the Unlock ( 🔓 ) position.

#### *Default Settings DIP Switch*

If the control settings have been erroneously set, the control can be made to operate with factory default values by switching the *Default settings* DIP switch. In order to override all current settings with the factory default values, the control must be powered down and powered up with the DIP switch set to *Default settings*. During normal operation set this DIP switch to *Run*.




#### *MODES 1*

The *MODES 1* adjustment selects the mode of operation for the mixing 1 section. There are three allowable settings:

• OFF - If the mixing 1 section is not used, the installer should select this setting. The control assumes the mixing 1 section is not used. All adjustments for the mixing 1 section will be skipped.




• RST (reset) - When using the mixing 1 section for building space heating, the installer should select this adjustment. The 201 control operates the mixing 1 section to provide outdoor reset. This is the factory setting.




• SETP (setpoint) - When the installer requires a fixed supply water temperature, this setting must be selected. The control operates the mixing 1 section to provide a fixed setpoint temperature equal to the *SETPNT 1* temperature adjustment.




#### *WWSD 1*

*WWSD 1* can be adjusted from 35 to 85°F, OFF (2 to 29°C). The mixing 1 section will shut down when the outdoor temperature is warmer than this setting. The *WWSD* setting also has an effect in the setpoint mode of operation. Factory setting is 68°F.




#### *SETPNT 1*

If setpoint mode of operation is selected, the control operates the variable speed injection pump 1 to provide a setpoint temperature equal to the *SETPNT 1* setting. The *SETPNT 1* setting can be adjusted from 70 to 200°F (21 to 93°C). An external mixing demand on the *Mix1 Demand* must be present. Factory setting is 110°F.




#### *SENSOR 1*

The *SENSOR 1* adjustment informs the control which external device is connected to the *Zo1 / Ind* terminal. There are three allowable settings:




• NONE - If no indoor sensor or tekmar zone control is connected to the *Zo1/Ind* terminal, this setting must be selected. This is the factory setting.




• INDR - If an indoor sensor is connected to the *Zo1/**Ind*** terminal, this setting must be selected. The control will provide indoor temperature feedback in order to fine tune the supply water temperature.




• ZOIN - If a tekmar Zone Control is connected to the ***Zo1***/*Ind* terminal, this setting must be selected. A tekmar Zone Control provides indoor temperature feedback from multiple zones.

#### *INDOOR 1*

The *INDOOR 1* setting can be adjusted from 35 to 95°F (2 to 35°C). This setting should be set to the highest desired indoor temperature in the mixing 1 section. Factory setting is 70°F.






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### DSGN RM1

This setting establishes the design room temperature on which the heat loss calculation was based. This setting is used by the control to calculate the heating curve for the mixing 1 section. The *DSGN RM 1* setting can be adjusted from 35 to 95°F (2 to 35°C). Factory setting is 70°F.




### DSN SUP 1

This setting establishes the design supply water temperature required in order to satisfy the heating system at the design outdoor temperature. This setting is used by the control to calculate the heating curve in the mixing 1 section. The *DSN SUP 1* setting can be adjusted from 80 to 220°F (27 to 104°C). Factory setting is 110°F.




### MAX SUP

The *MAX SUP 1* adjustment is used to set the highest allowable temperature in the mixing 1 section. This setting can be adjusted from 100 to 220°F (38 to 104°C). Factory setting is 180°F.




### OUT DSGN

The outdoor design temperature is obtained from local weather data. This setting can be adjusted from -50 to 30°F (-45 to -1°C). Factory setting is 10°F.




### MODES 2

The *MODES 2* adjustment selects the mode of operation for the mixing 2 section. There are four allowable Settings:




- OFF - If the mixing 2 section is not used, the installer should select this setting. The control assumes the mixing 2 section is not used. All adjustments for the mixing 2 section will be skipped.




- RST (reset) - When using the mixing 2 section for building space heating, the installer should select this adjustment. The 201 control operates the mixing 2 section to provide outdoor reset. This is the factory setting.




- SETP (setpoint) - When the installer requires a fixed supply water temperature, this setting must be selected. The control operates the mixing 2 section to provide a fixed setpoint temperature equal to the *SETPNT 2* temperature adjustment.




- SNOW (melting) - If section 2 is used for a snow melting application, the installer must select this setting. The control then operates the mixing 2 section to regulate an outdoor slab.

### WWSD 2

*WWSD 2* can be adjusted from 35 to 85°F, OFF (2 to 29°C). The mixing 2 section will shut down when the outdoor temperature is warmer than this setting. The *WWSD* setting also has an effect in the setpoint mode of operation. Factory setting is 68°F.




### SETPNT 2

If setpoint mode of operation is selected, the control operates the variable speed injection pump 2 to provide a setpoint temperature equal to the *SETPNT 2* setting. The *SETPNT 2* setting can be adjusted from 70 to 200°F (21 to 93°C). An external mixing demand on the *Mix2 Demand* must be present. Factory setting is 110°F.




### SENSOR 2

The *SENSOR 2* adjustment informs the control which external device is connected to the *Zo2 /Ind* terminal. There are three allowable settings:




- NONE - If no indoor sensor or tekmar zone control is connected to the *Zo2/Ind* terminal, this setting must be selected. This is the factory setting.




- INDR - If an indoor sensor is connected to the *Zo2 /**Ind*** terminal, this setting must be selected. The control will provide indoor temperature feedback in order to fine tune the supply water temperature.




- ZOIN - If a tekmar Zone Control is connected to the ***Zo2*** */Ind* terminal, this setting must be selected. A tekmar Zone Control provides indoor temperature feedback from multiple zones.

### INDOOR 2

The *INDOOR 2* setting can be adjusted from 35 to 95°F (2 to 35°C). This setting should be set to the highest desired indoor temperature in the mixing 2 section. Factory setting is 70°F.






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### *DSGN RM 2*

This setting establishes the design room temperature on which the heat loss calculation was based. This setting is used by the control to calculate the heating curve for the mixing 2 section. The *DSGN RM 2* setting can be adjusted from 35 to 95°F (2 to 35°C). Factory setting is 70°F.




### *DSN SUP 2*

This setting establishes the design supply water temperature required in order to satisfy the heating system at the design outdoor temperature. This setting is used by the control to calculate the heating curve for the mixing 2 section. The *DSN SUP 2* setting can be adjusted from 80 to 220°F (27 to 104°C). Factory setting is 110°F.




### *MAX SUP*

The *MAX SUP 2* adjustment is used to set the highest allowable temperature in the mixing 2 section. This setting can be adjusted from 100 to 220°F (38 to 104°C). Factory setting is 180°F.




### *MELTING*

The *MELTING* setting is the desired slab temperature when snow melting is required. It can be adjusted from 32 to 77°F (0 to 25°C). Factory setting is 36°F.




### *IDLING*

The *IDLING* setting is the desired slab temperature during idling, and it can be adjusted from 20 to 68°F (-7 to 20°C). If Idling is not desired, this setting must be set to *OFF*.




### *ΔT MAX*

The 201 control limits the rate of heat applied to a slab to prevent excessive thermal stress. The rate of heat delivery to the slab is dependant on the difference between the melting system supply and return temperatures. The control measures these temperatures and calculates the difference to give the ΔT. If this calculated ΔT approaches the *ΔT MAX* setting, the control operates the variable speed injection pump to maintain the ΔT at the *ΔT MAX* setting. The *ΔT MAX* setting can be adjusted from 10 to 70°F (5 to 39°C). Factory setting is 30°F.



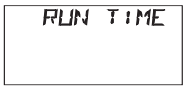

### *CWCO (Cold Weather Cut Off)*

*CWCO* is the lowest temperature at which the melting system continues to operate. This temperature is set based on the capacity of the snow melt system and the economics of melting in extreme conditions. The range of this setting is from -20 to 50°F (-29 to 10°C). The *CWCO* adjustment has an *OFF* position for systems where a minimum outdoor temperature is not a consideration.




### *RUN TIME*

The *RUN TIME* is set to the estimated time required to complete snow melting. This setting may be adjusted from 30 min. to 18 hours. For continuous operation the *RUN TIME* may be set to *INF* (infinite). Factory setting is 4 hours. The snow melting run time will start counting down once the slab is up to *MELTING* temperature.




### *BOILER RETURN*

To prevent problems of flue gas condensation in the boiler, the *Boil RET* adjustment is set to the minimum boiler return temperature as specified by the boiler manufacturer. When the boiler return approaches this setting, the control operates the variable speed pump to limit the output until the boiler return can warm up. If a minimum boiler return temperature is not required, as with condensing or electric boilers, this adjustment can be set to *OFF*. This setting can be set from 70 to 180°F (21 to 82°C). Factory setting is 130°F.




### *UNITS (°F - °C)*

The control can display temperatures in °C or °F. Factory setting is °F.

### *CONTRAST*

In order to improve visibility of the Liquid Crystal Display (LCD) the *CONTRAST* of the display may be adjusted.

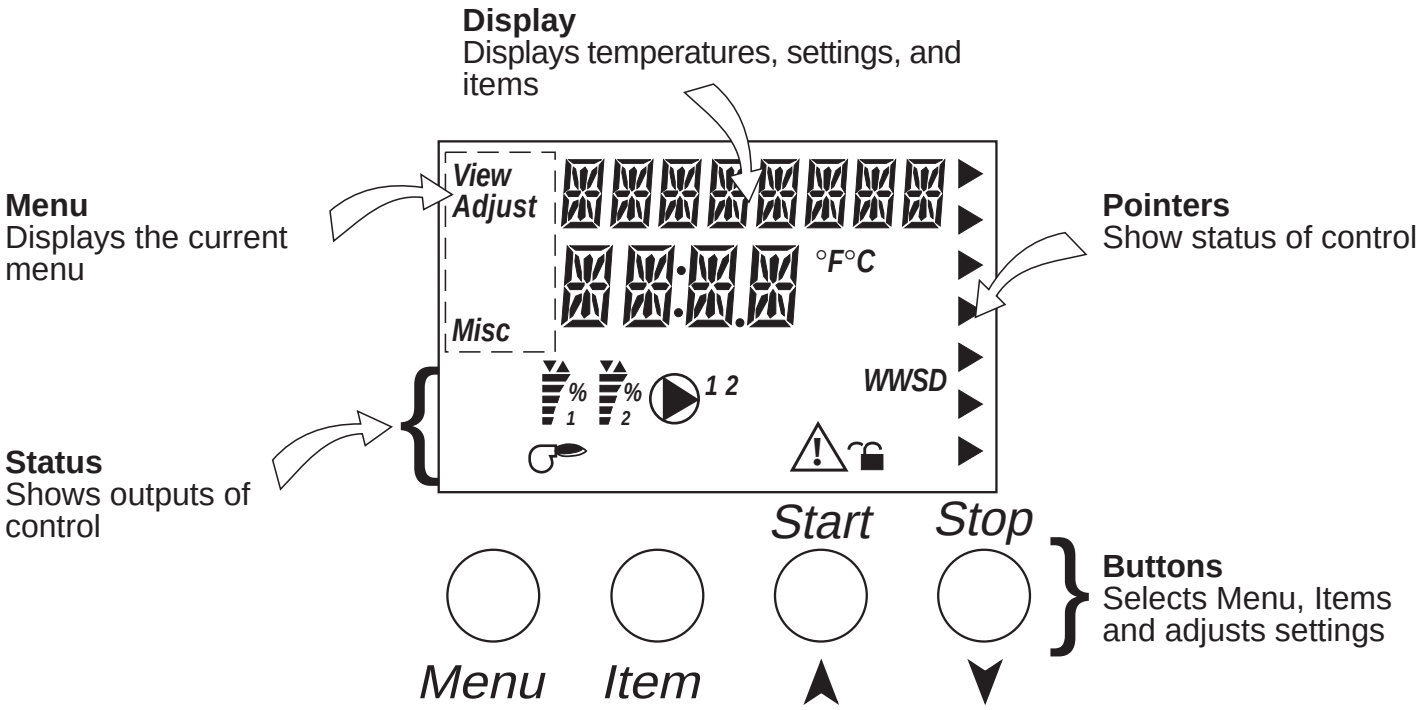
### *BACKLITE*

The *BACKLITE* on the display may be adjusted to OFF, 20 S, or ON. When it is adjusted to 20 S the display is illuminated for 20 seconds after a button is pressed.



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## Description of Display Elements




| | | | |
|---|---|---|---|
| % 1 % 2 | **Variable speed output scale**<br>Shows the speed of the injection pump1 or 2. Arrows show whether pumps are speeding up or slowing down | ⚠ | **Warning**<br>Displays when an error is present |
| 1 2 | **Pumps**<br>Displays when mixing pumps 1 or 2 are on. | 🔓 | **Lock-Unlock**<br>Displays whether keypad is locked or unlocked |
| | **Burner**<br>Display when the Boiler Enable contact is closed. | WWSD | **WWSD**<br>Displays when control is in Warm Weather Shut Down |



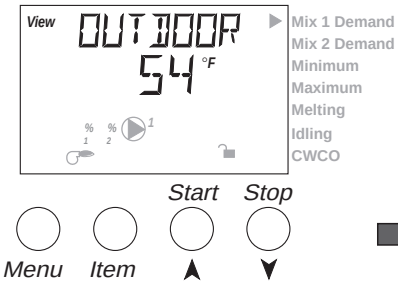
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# View Menu

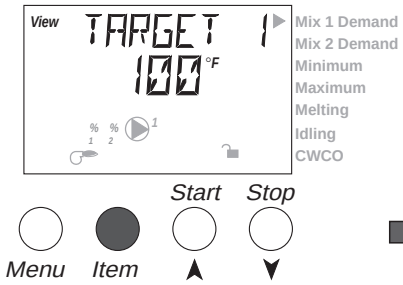
## RST (RESET) AND SETP (SETPOINT) MODE

To view the different items, continuously press the *Item* button.

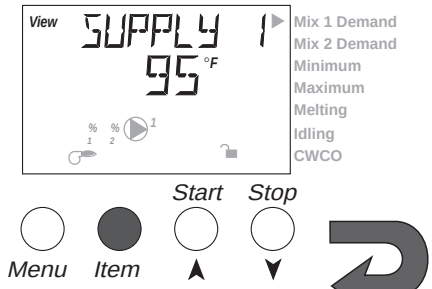
**Outdoor temperature**




**Target Mixing 1 Temperature**



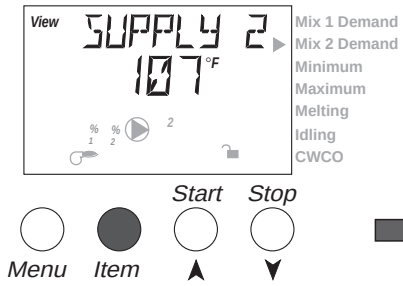

**Actual Mixing 1 Supply Temperature**




**Target Mixing 2 Temperature**




**Actual Mixing 2 Supply Temperature**




**Boiler Return Temperature**



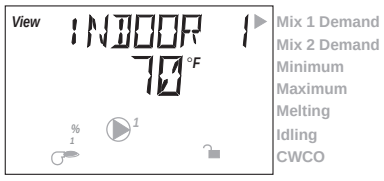

- If a mixing section is turned off, the ***View Items*** from that mixing section are skipped.
- Dashes on the *TARGET* item denote heat is not required in that mixing section.
- Continue to press the ***Item*** button to toggle between view items.
- When attempting to see a given temperature and a sensor fault is present the control displays "**Err**".

**Target Mixing 2 Temperature**



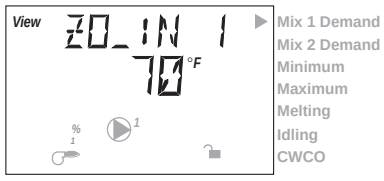

## INDOOR SENSOR / ZONE CONTROL

**Actual Section 1 indoor Temperature**




OR

**Zone Control Target Temperature**




- If an indoor sensor is selected in the *Adjust* menu, the actual indoor temperature may be viewed.
- If a zone control is selected in the adjust menu, the zone control target temperature may be viewed.

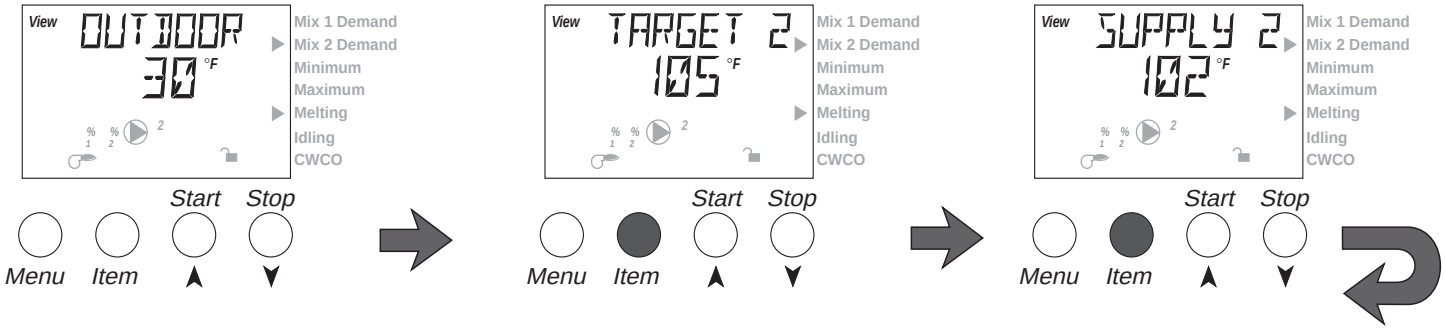


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## SNOW (SNOWMELTING) MODE

This display assumes that the Mixing 1 section is turned off and therefore no items for that section is present.

### Outdoor temperature



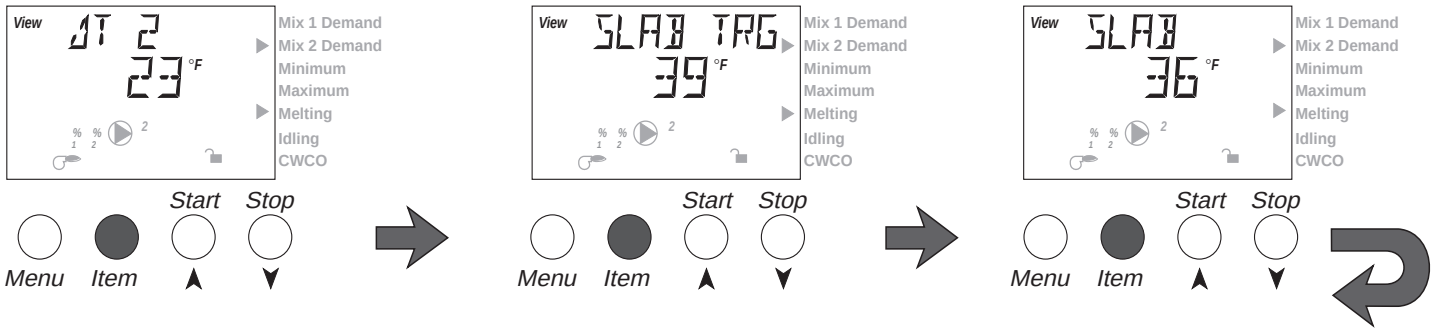

### Target Mixing 2 Temperature




### Actual Mixing 2 Supply Temperature




### ΔT 2 (Temperature difference)



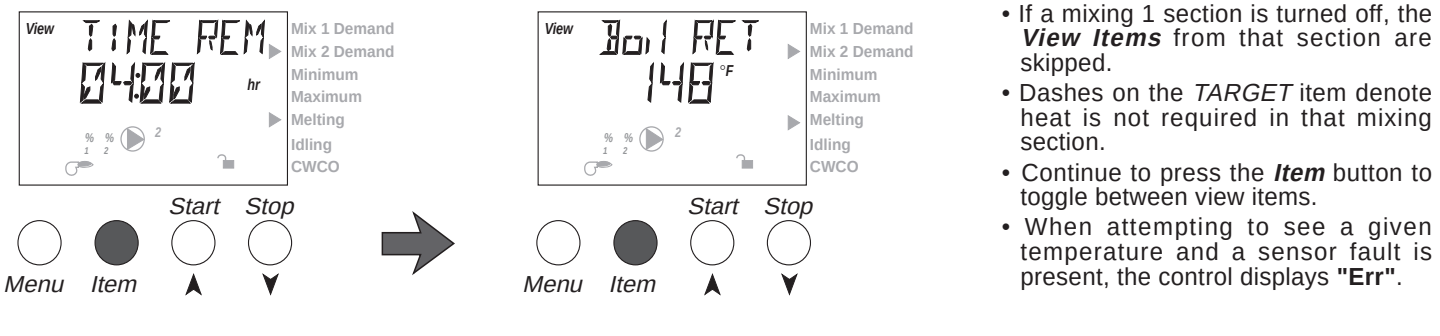

### Slab Target Temperature




### Actual Slab Temperature




### Time Remaining




### Boiler Return Temperature




- If a mixing 1 section is turned off, the ***View Items*** from that section are skipped.
- Dashes on the *TARGET* item denote heat is not required in that mixing section.
- Continue to press the ***Item*** button to toggle between view items.
- When attempting to see a given temperature and a sensor fault is present, the control displays **"Err"**.

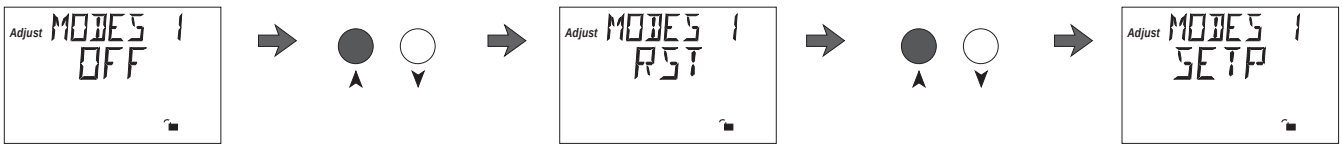
## Adjust Menu




**To enter the *Adjust* menu press the *Menu* button.**

## MIXING 1 SECTION

### Mixing 1 Section Off




### Mixing 1 Section in Reset Mode



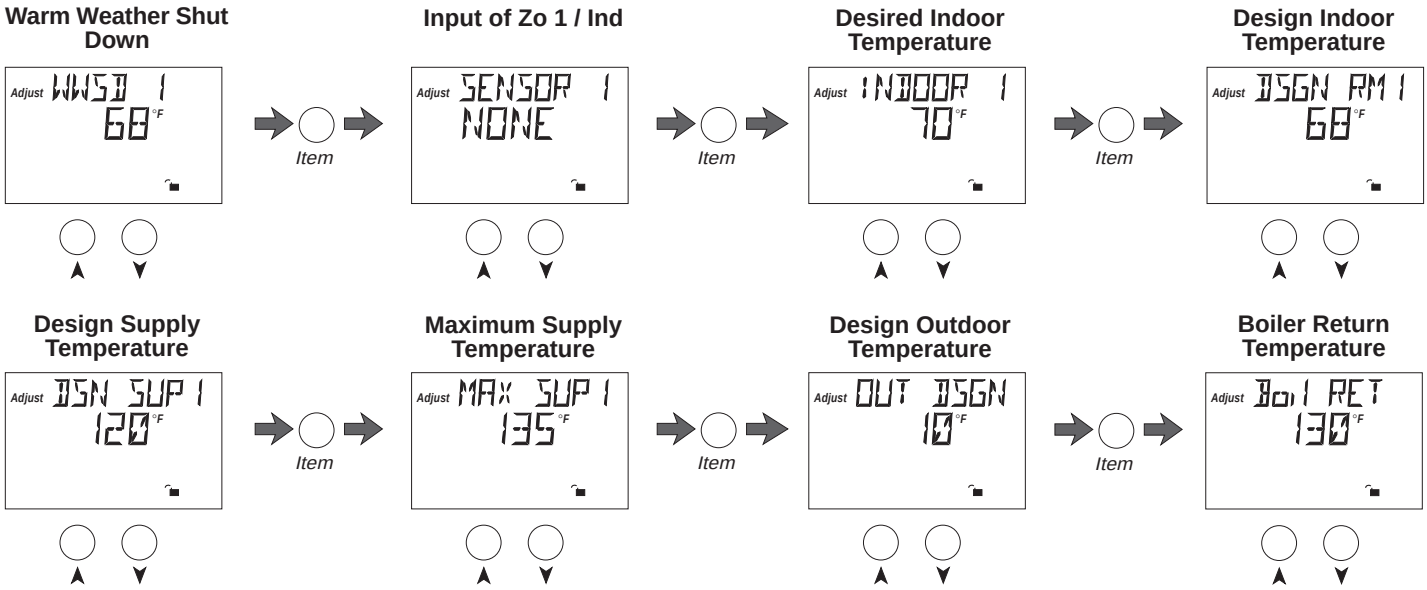

### Mixing 1 Section in Setpoint Mode



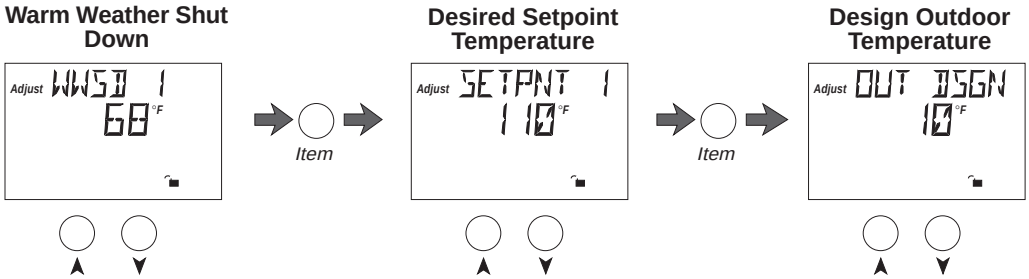



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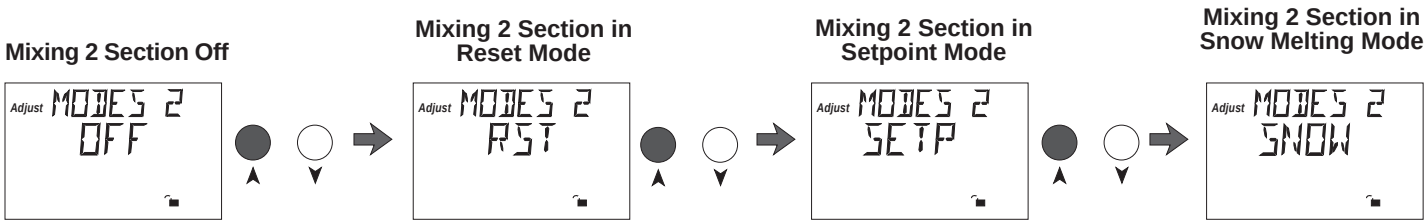
## MIXING 1 IN RESET MODE



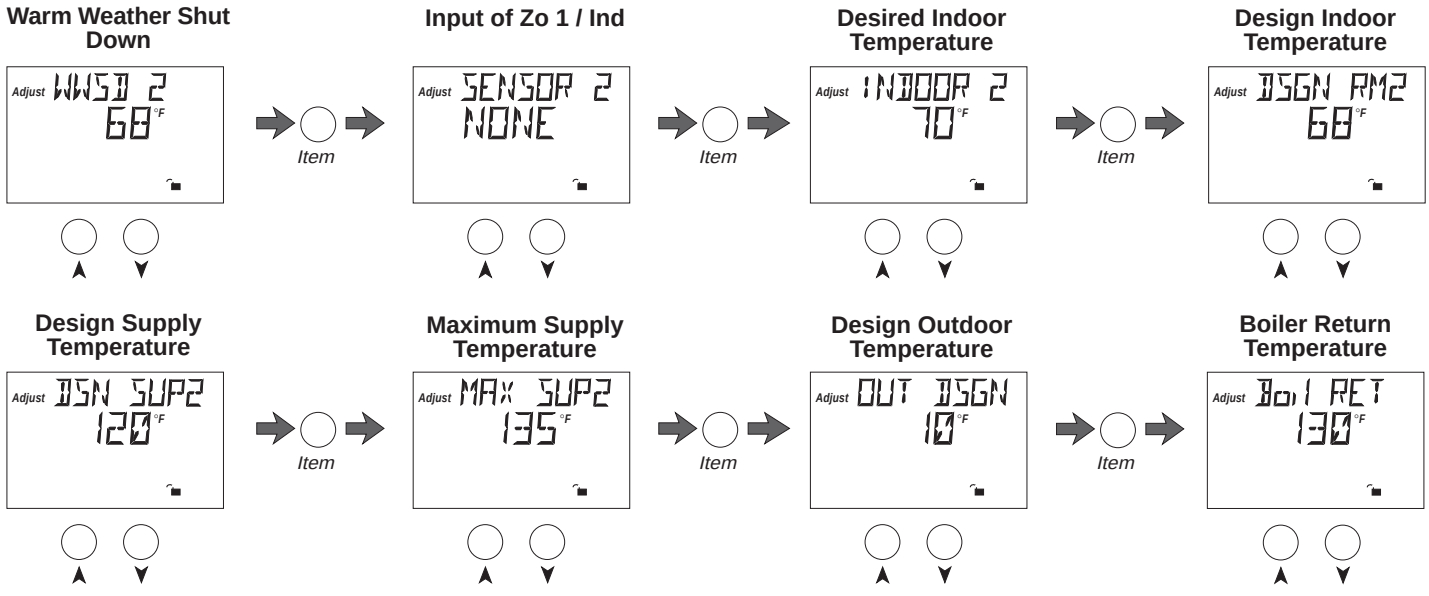

## MIXING 1 IN SETPOINT MODE




## MIXING 2 SECTION




## MIXING 2 IN RESET MODE

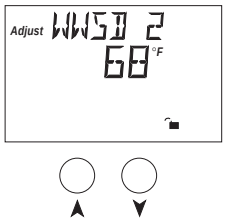





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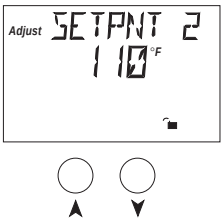
## MIXING 2 IN SETPOINT MODE

**Warm Weather Shut Down**



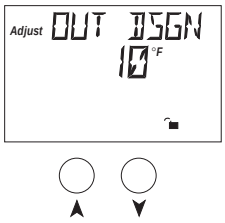

Item

**Desired Setpoint Temperature**



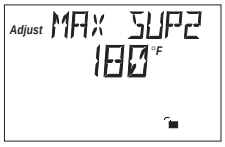

Item

**Design Outdoor Temperature**



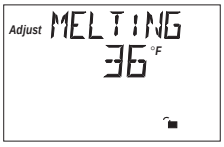

## MIXING 2 IN SNOW MELTING MODE

**Maximum Supply Temperature**



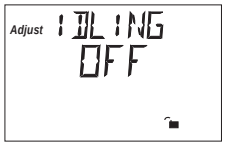

Item

**Slab Melting Temperature**



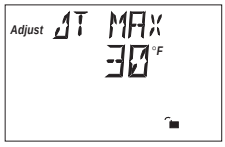

Item

**Slab Idling Temperature**

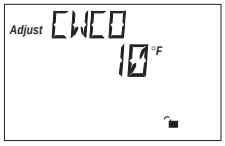



Item

**Maximum ΔT**

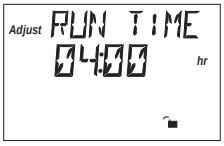



**Cold Weather Cut Off**



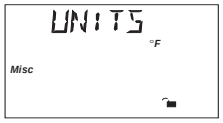

Item

**Melting Run Time**



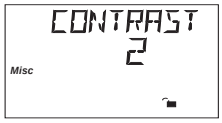

## Misc. Menu

**Units (°F - °C)**



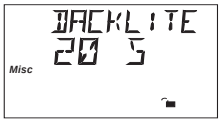

Item

**Display Contrast**




Item

**Display Backlight**




THE CONTROL AUTOMATICALLY GOES BACK TO THE VIEW MENU WHEN THE BUTTONS ARE LEFT ALONE FOR 20 SECONDS



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## Display Menu (Cont.)

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## Testing the Control

### STEP SEVEN — OPERATIONAL TEST OF CONTROL FUNCTIONS

The Wirsbo DuoMix 201 has a test routine which is used to test the main control functions. The 201 continually checks the sensors and displays an error message whenever a fault is found. See page 20 for the list of error messages. When the *Test* button is pushed, the *WWSD* display and the Minimum, Maximum, Melting, Idling, and CWCO pointers are turned off. The individual outputs and relays are then tested in the following test sequence.

#### *Test Sequence*

Each step in the test sequence lasts 10 seconds. During the test routine, given a Mix demand is present, the test sequence can be paused by pressing the *Test* button. Once the test sequence is paused, the *Test* button can be pressed again to skip to the next step. This can also be used to rapidly skip through the test sequence. To reach the desired step, repeatedly press and release the *Test* button until the appropriate device and indicator light turn on.

*STEP 1* - The variable speed injection pump 1 (*Var 1 Pmp*) is ramped up to 100% over 10 seconds. If the mixing 1 section is not used (i.e. *MODES 1* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 2* - The variable speed injection pump 1 (*Var 1 Pmp*) is ramped down to 0% over 10 seconds. If the mixing 1 section is not used (i.e. *MODES 1* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 3* - The mixing 1 pump (*Mix1 Pmp*) is turned on for 10 seconds. If the mixing 1 section is not used (i.e.. *MODES 1* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 4* - The variable speed injection pump 2 (*Var 2 Pmp*) is ramped up to 100% over 10 seconds. If the mixing 2 section is not used (i.e. *MODES 2* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 5* - The variable speed injection pump 2 (*Var 2 Pmp*) is ramped down to 0% over 10 seconds. If the mixing 2 section is not used (i.e. *MODES 2* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 6* - The mixing 2 pump (*Mix2 Pmp*) is turned on for 10 seconds. If the mixing 2 section is not used (i.e. *MODES 2* set to *OFF*) *N/A* is displayed and the pump is not operated.

*STEP 7* - The *Boiler Enable* relay is turned on for 10 seconds.

#### *Pointers "On"*

| | |
|---|---|
| *Mix 1 Demand* | • Mixing 1 section requires heat. |
| *Mix 2 Demand* | • Mixing 2 section requires heat. |
| *Minimum* | • The return temperature to the boiler is below the *Boil RET* setting and the 201 is reducing the load in order to increase the boiler temperature as fast as possible. |
| *Maximum* | • The supply temperature of mixing section 1 or 2 is approaching the *Max Sup* setting and the 201 will not increase the supply temperature any further. |
| *Melting* | • The control is in melting mode. |
| *Idling* | • The control is in Idling mode. |
| *CWCO* | • The control is in Cold Weather Cut Off and heat for snow melting is not provided, unless the slab is already up to temperature. |

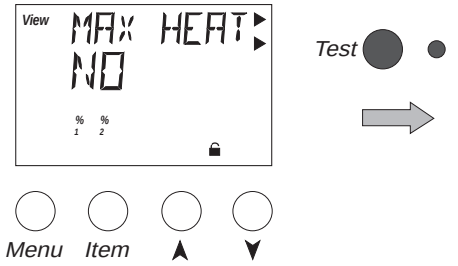
#### *MAX HEAT Function*

The *MAX HEAT* function provides maximum heat to the system by targeting the maximum supply temperature for each mixing section. This function is useful during system start up, manual override, or testing during the summer. When the control is in *MAX HEAT* and a mixing demand in the mixing section 1 is present, the control turns on the mixing pump relay (*Mix1 Pmp*), the variable speed pump output (*Var1 Pmp*) and *Boiler Enable* relay. The *MAX HEAT* function overrides the WWSD and the boiler minimum return temperature, but it will not exceed the maximum supply setting. When a mixing demand in the mixing 2 section is present, the control operates similarly as explained above. The speed of the injection pump(s) may be adjusted using the arrow keys to limit the amount of injection desired. Once enabled, the *MAX HEAT* function operates for 24 hours or until any of the *Test*, *Menu*, or *Item* buttons are pressed. See next page for setup instructions.

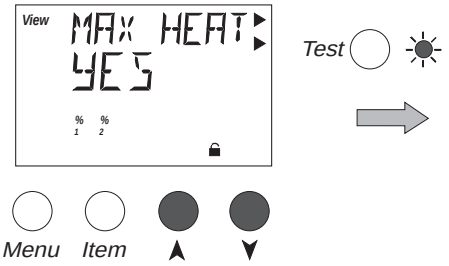


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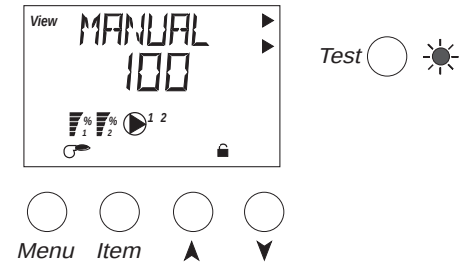
Press and Hold the **Test** button for 3 seconds



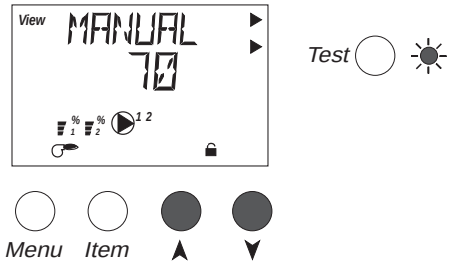

Comfirm if *MAX HEAT* is desired by pressing the **up** or **down** button



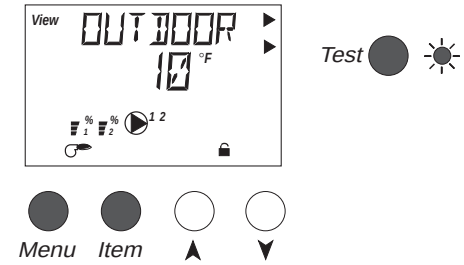

Control displays MANUAL and the % speed of the variable speed pump(s)




The % speed output may be adjusted using the **up** and **down** buttons




To exit the *MAX HEAT* function presseither the ***Test***, ***Menu*** or ***Item*** buttons




## STEP EIGHT — TROUBLESHOOTING

As in any troubleshooting procedure, it is important to isolate a problem as much as possible before proceeding. The Error Messages and *Test* button greatly simplify troubleshooting of the 201. When the control is displaying an Error Message, identify the fault from the look-up table on page 20 and follow standard testing procedures to confirm the problem. If you suspect a wiring fault, return to steps three, four and five, and carefully check all external wiring and wiring connections.

### *Sensor and Internal Faults*

Whenever a fault is detected in any of the sensors, an error message will be displayed in the LCD.

### *Outdoor Sensor Fault*

When the outdoor sensor is open circuited or missing, the control will display *OUTDOOR OPEN*. If a short circuit occurs, the control displays *OUTDOOR SHRT*. When a fault with the outdoor sensor is present, the control assumes a fixed outdoor temperature of 32°F. During snow melting operation, the control assumes an outdoor temperature of 20°F.

### *Boiler Return Sensor Fault*

When the Boiler Return Sensor is open circuited or missing and the *Boil RET* adjustment is not set to *OFF*), the control will display *Boil RET OPEN*. If a short circuit occurs, the control displays *Boil RET SHRT*. When a fault with the boiler return sensor is present, the control continues to operate as if no return sensor is installed. If the system is providing snow melting, the mixing 2 section is turned off.

### *Supply 1 Sensor Fault*

When the Supply Sensor of the mixing 1 section is open circuited or missing and *MODE 1* is not set to *OFF*, the control will display *SUPPLY 1 OPEN*. If a short circuit occurs, the control displays *SUPPLY 1 SHRT*. When a fault with the Supply Sensor is present, the control shuts down the mixing 1 section.

### *Supply 2 Sensor Fault*

Operates in a similar manner as the mixing 1 section.

### *IND / Zo 1*

If *ZOIN* or *INDR* is selected and an open circuit is seen on the Zo 1/Ind terminal, the control will display *IND/Zo 1 OPEN*. If a short circuit occurs, the control displays *IND/Zo 1 SHRT*. When a fault is present and *ZOIN* is selected, the control ignores any demands from the zone control. If *INDR* is selected, the control provides outdoor reset but no indoor temperature feedback.

### *IND / Zo 2*

Operates in a similar manner as the mixing 1 section.



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### *Return 2 Sensor Fault*

When the mixing 2 section is set to *SNOW* and a Return Sensor open circuit occurs, the control will display *RETURN 2 OPEN*. If a short circuit occurs, the control displays *RETURN 2 SHRT*. When a fault with the Return 2 Sensor is present, the control continues to do snow melting.

### *Slab Sensor Fault*

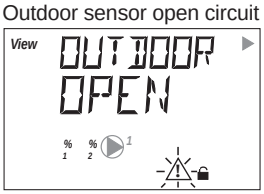
When the Slab Sensor is open circuited or missing during snow melting operation, the control will display *SLAB OPEN*. If a short circuit occurs, the control displays *SLAB SHRT*. When a fault with the Slab sensor is present, the control shuts down the snow melting system.

### *tN2 Error*

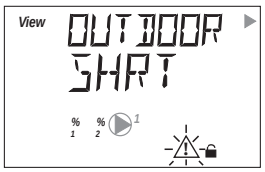
When the wires on the tN2 device (Remote Display Module 040 or Snow melt Enable Device 038) are reversed or the contacts are shorted the control displays *tN2 SHRT* and continues normal operation.

## Error Messages

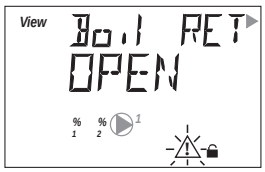
Outdoor sensor open circuit




Outdoor sensor short circuit



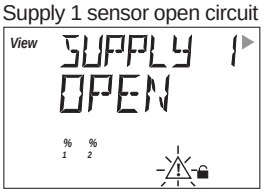

Boiler return sensor open circuit



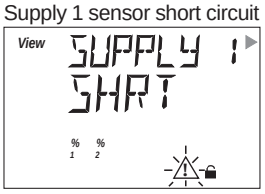

Boiler return short circuit



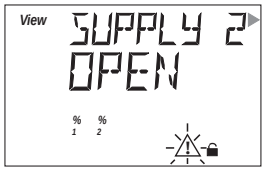

Supply 1 sensor open circuit



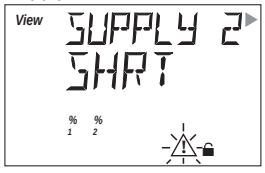

Supply 1 sensor short circuit



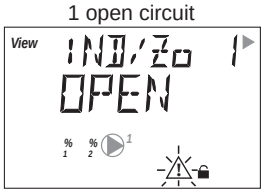

Supply 2 sensor open circuit



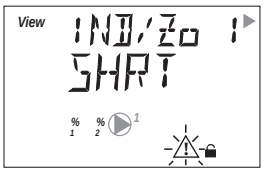

Supply 2 sensor short circuit



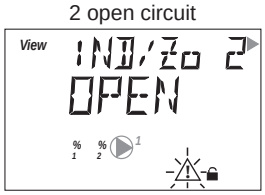

Indoor sensor/zone control 1 open circuit




Indoor sensor/zone control 1 short circuit



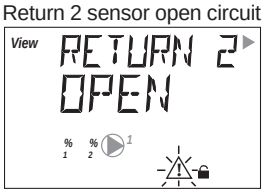

Indoor sensor/zone control 2 open circuit



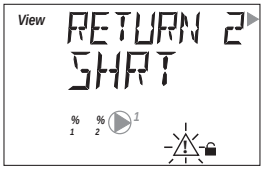

Indoor sensor/zone control 2 short circuit



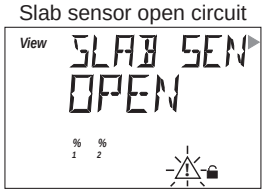

Return 2 sensor open circuit



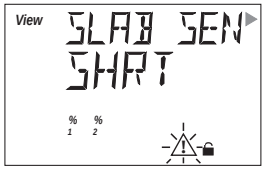

Return 2 sensor short circuit



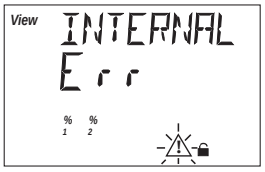

Slab sensor open circuit




Slab sensor short circuit




Internal error




tN 2 error

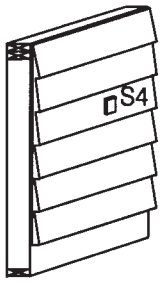



**REFER TO THE TROUBLESHOOTING SECTION ABOVE FOR OPERATION DETAILS**

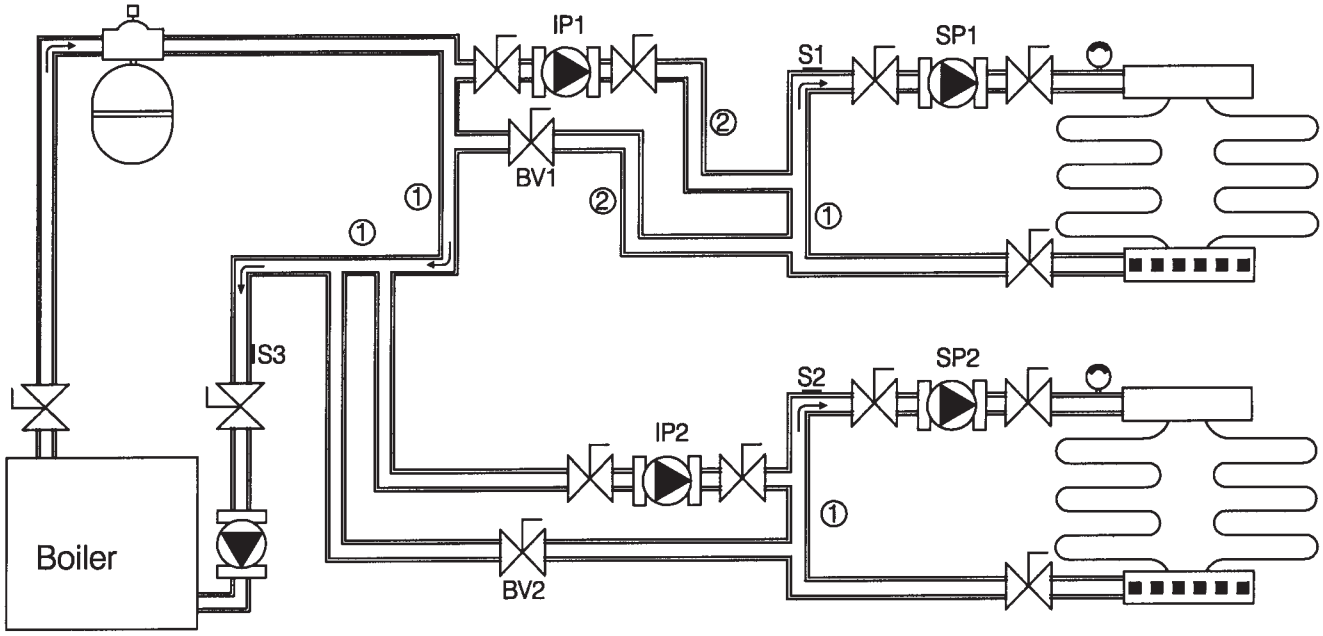


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## Boiler
Injection Pumping
- Dual Water Temperature




PLACE OUTDOOR SENSOR
ABOVE SNOW LINE
ON NORTH FACING WALL



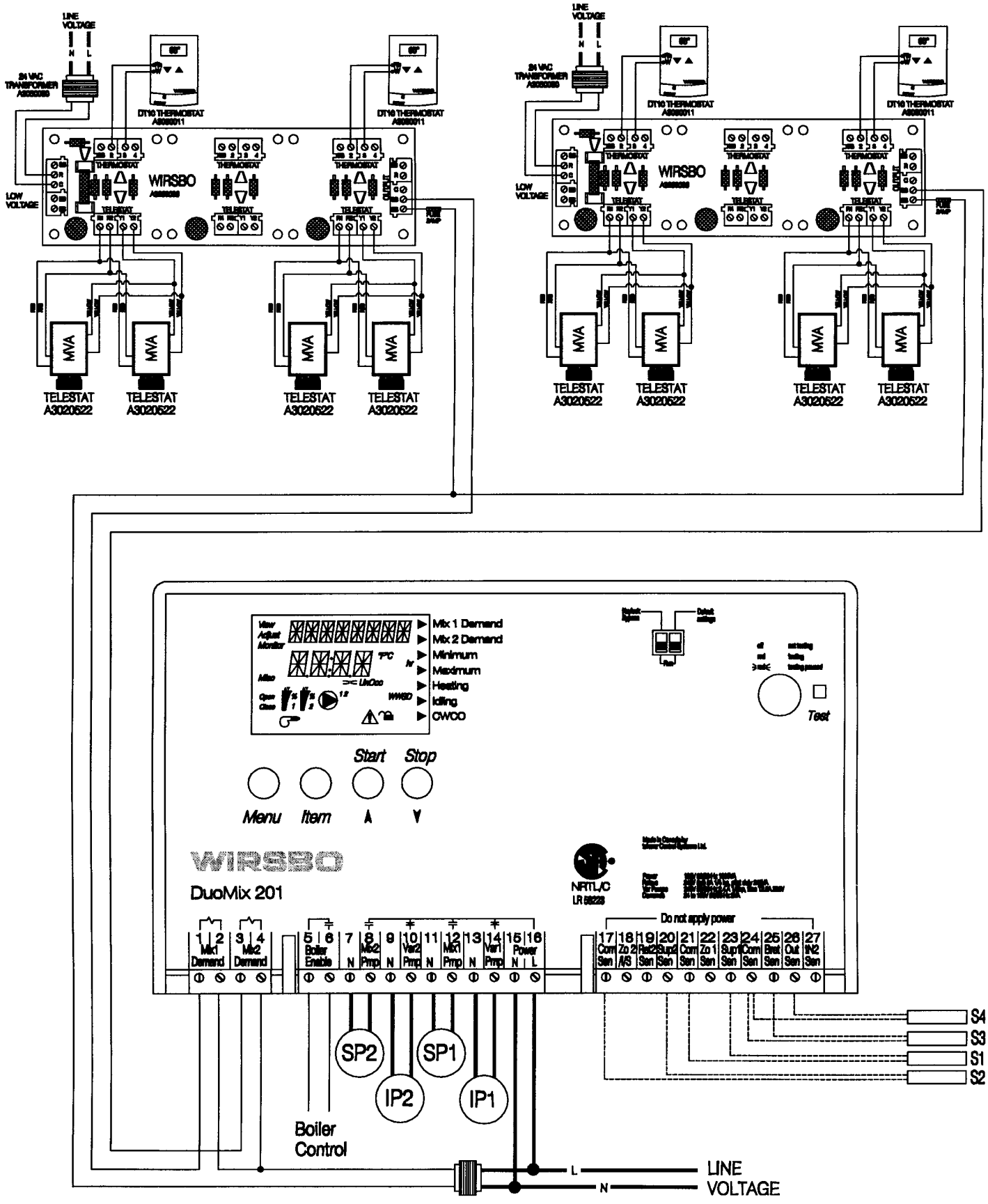

Note: ① Spacing between tees not to exceed 6"
② Thermal Trap - minimum 12"

## Legend

- Radiant Panel
- Manifold w/telestats
- Manifold w/valves
- Manifold wo/valves
- Circulator
- Tempering Valve
- Expansion Tank
- Baseboard
- Zone Valve
- Ball Valve
- Flow Control
- Temperature Gauge
- Directional Flow



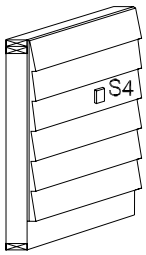
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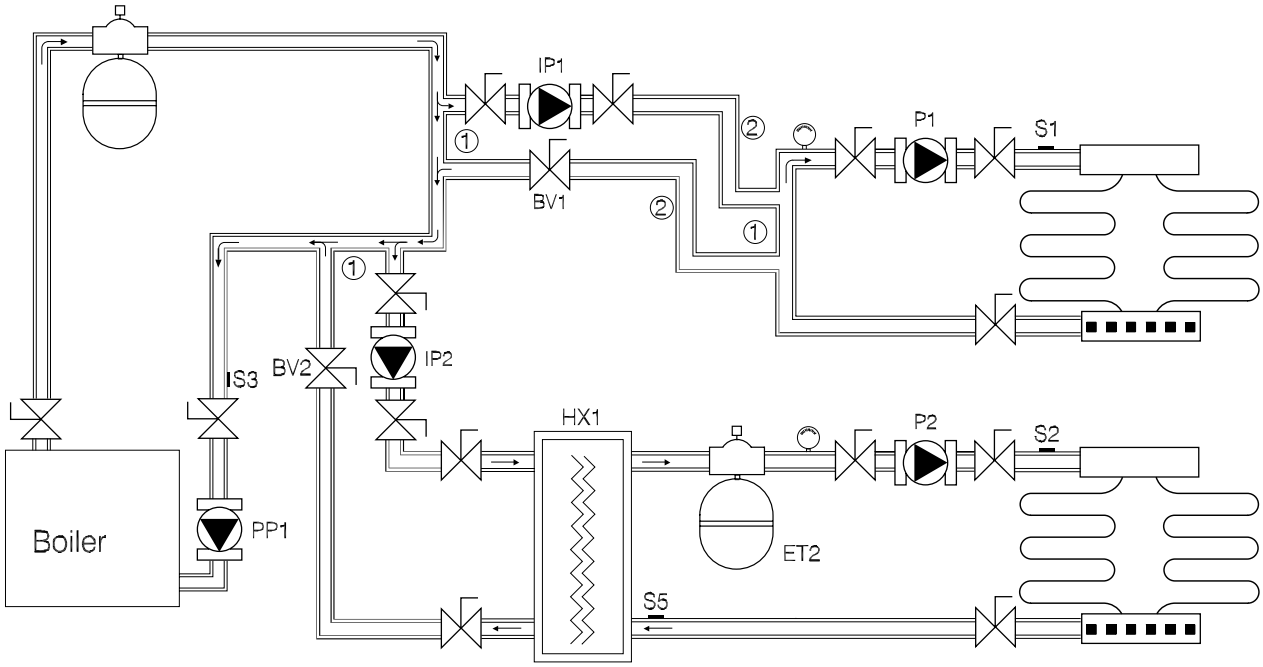



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## Non-Condensing Boiler
Single Temperature RFH & Snow Melt
Wirsbo DuoMix 201




PLACE OUTDOOR SENSOR
ABOVE SNOW LINE
ON NORTH FACING WALL



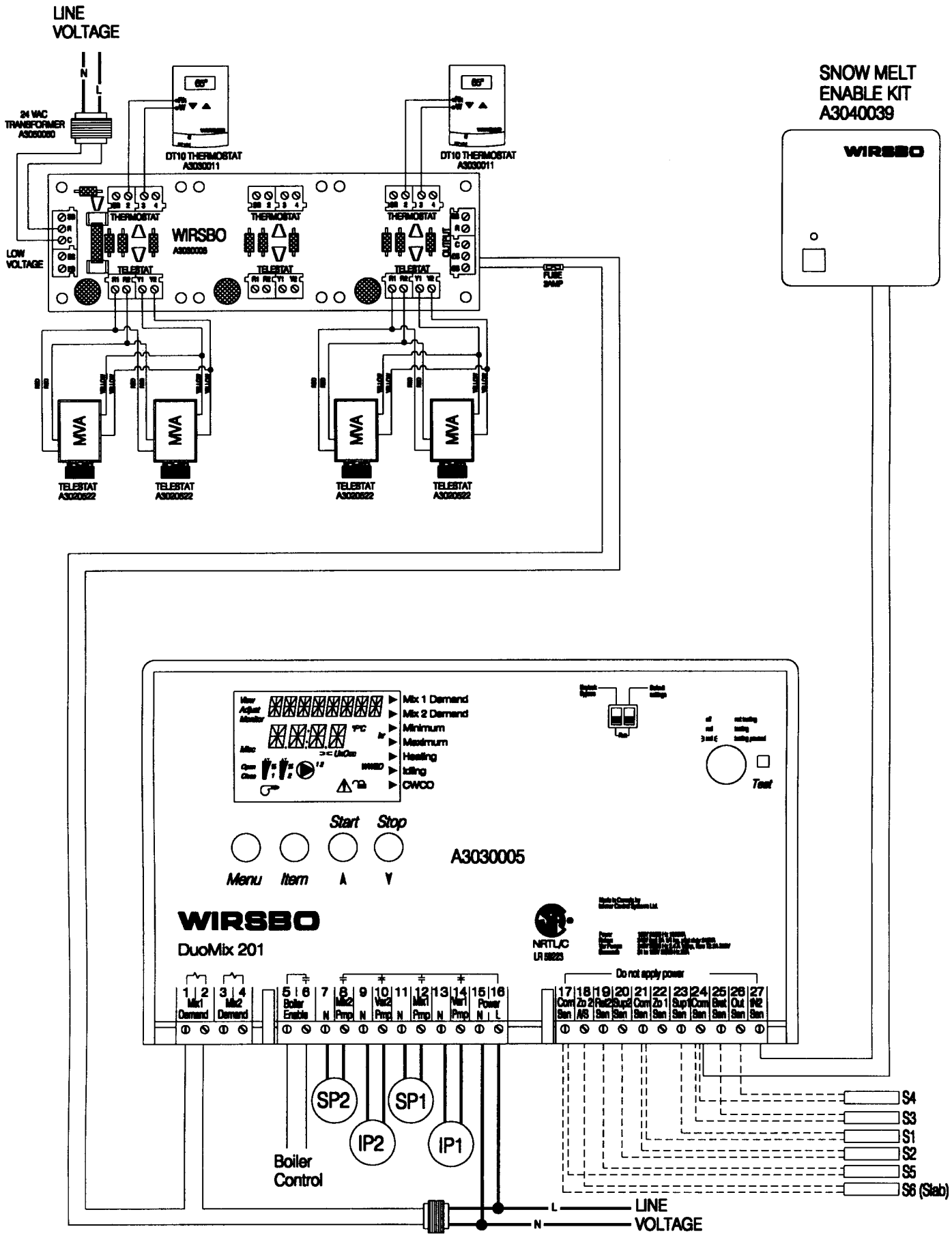

Note: ① Spacing between tees not to exceed 6"
② Thermal trap - minimum 12"

## Legend

- Radiant Panel
- Manifold w/telestats
- Manifold w/valves
- Manifold wo/valves
- Circulator
- Tempering Valve
- Expansion Tank
- Baseboard
- Zone Valve
- Ball Valve
- Flow Control
- Temperature Gauge
- Directional Flow



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## Variable Speed Injection Mixing For Hydronic Heating Systems*

The purpose of this essay is to explore the practical application of Variable Speed Injection Mumping to effect precise heat transfer from high temperature to low temperature hydronic heating system loops. Often the available heat source produces higher temperature water than is required by the heat delivery system. To effectively achieve this lower water temperature, the high temperature water must be injected into the system at a controlled rate to mix with the system return water, raising the temperature just enough to meet the heating load. Various devices and plumbing arrangements can be used to accomplish this.

In some instances, a heat source - condensing or electric boiler, geo-thermal heat pump, etc. - can be operated at low temperatures and dedicated solely to operating a low temperature radiant heating system. In the vast majority of systems, however, mixing is required because:

- A boiler minimum operating temperature is required
- High temperature water is required for other system needs
- Water temperatures vary over a wide range, i.e. solar heat sources, waste heat utilization, wood fired boilers, etc.

When the water temperatures of both the high and low temperature loops are constant, with a constant heating load, a simple manual bypass balancing arrangement (Figure 1) can be used to maintain a steady water temperature since it continually injects high temperature water at a fixed rate.

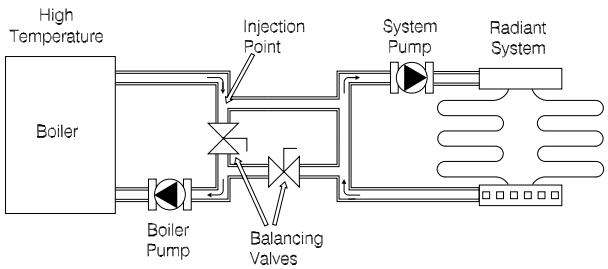



Figure 1 Mixing with fixed balancing valves

*Text and diagrams used with permission from tekmar

When the high temperature supply fluctuates, and/or varying water temperatures are required in the heating system at a constant load, an on/off valve or pump (Figure 2) can be used to intermittently inject hot water at a fixed rate whenever heat is required.

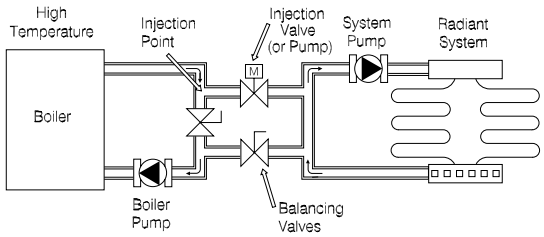



Figure 2 Mixing with on/off injection valve or pump

In most heating systems, the high temperature supply fluctuates, the system loop temperature requirements constantly change and the heating load constantly changes. These factors, combined with the common need to maintain a safe minimum boiler return temperature, require that the rate of heat injection be precisely controlled and constantly readjusted.

The easiest, most reliable and most cost effective way to accomplish this is to use Variable Speed Injection Mixing with the Wirsbo DuoMix 201 System Controller.

Variable Speed Injection Mixing (Figure 3) can be used to transfer heat from a Primary, high temperature boiler loop to a Secondary, low temperature heating loop. The speed of the Injection Circulator is automatically adjusted to deliver the desired volume of hot water to the low temperature loop. Constant flow can be maintained in both loops regardless of the circulator speed, and more or less heat is injected as the circulator speed is increased or decreased. The circulator speed is constantly adjusted as the heating system load and the supply water temperature change. If the boiler return temperature becomes too cold, the circulator can be slowed down to reduce the heat injection rate, resulting in an increased boiler return temperature.



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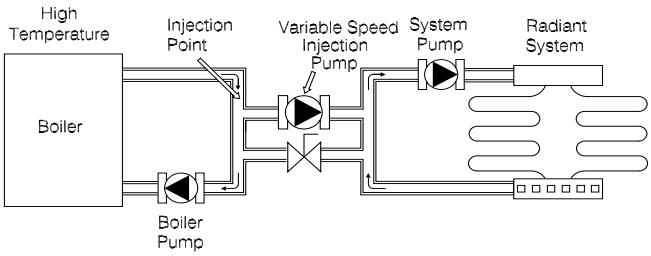



Figure 3 Mixing with variable speed pump

tekmar Control Systems has developed a new series of controls that utilizes variable speed circulator output. For residential and small commercial systems, the controls have a 120Vac 50/60Hz output to directly power small circulators. This output modulates the power supply to the circulator to vary its its rotational speed. The maximum allowable amperage for this output is 2.2 Amps, which limits the allowable circulator size to ⅙ HP.

## Pump Types

The 120Vac output of the control can operate a variety of small circulators produced by various manufacturers. A permanent capacitor, impedance protected motor (no start switch) under ⅙ HP is required. Most small "wet rotor" circulators have proven to acceptable.

## Plumbing Arrangements

Variable Speed Injection Mixing requires water from the boiler - or Primary - loop to be injected into the low temperature - or Secondary - loop so that the heat rise and the mixing occur directly after the point of injection, downstream of the return to the hot loop. (See Figures 4&5).

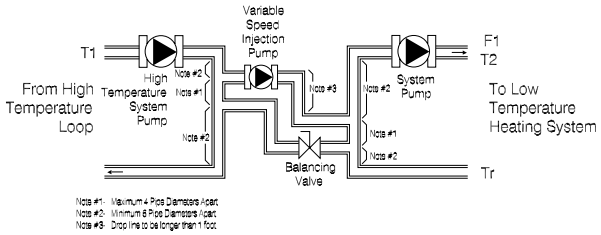



Figure 4 Injection into a horizontal loop

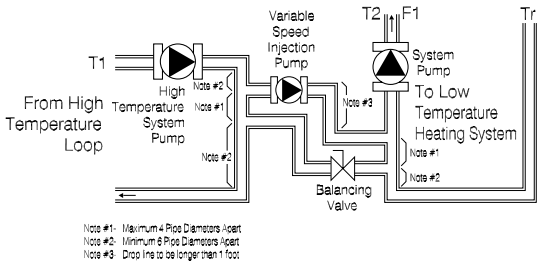



Figure 5 Injection into a vertical loop

## Advantages

This type of system can use a small circulator to inject a high BTU input into a relatively large system flow. Typically, the injection pump need only deliver ⅛ to ¼ of the system flow for low temperature radiant panels if high temperature water is available for injection.

## Disadvantages

In small systems, the smallest available circulator for variable speed injection may be too large. Check circulator sizing carefully.

## Injection Pump Sizing

Figure 6 illustrates the mixing action and cross-over flow that occurs in direct injection systems. For proper sizing, the designer must know:

$F_1$ - System Supply Flow Rate in GPM

$T_1$ - Hot - or Primary - Loop Supply Temperature Available

$T_2$ -Low Temperature - or System - Supply Temperature

$T_R$ - Low Temperature - or System- Return Temperature

$\Delta T_S$ - Low Temperature - or System - Temperature Differential ($T_2$ - $T_R$)

Note : All values are to be given at design conditions.

The formula that expresses the temperature vs. flow relationship that is illustrated in Figure 6 is:

$$F_V = \frac{F_1 \times \Delta T_S}{T_1 - T_R}$$



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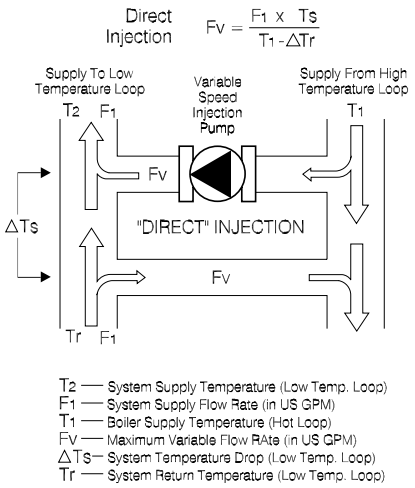



Figure 6 Direct Injection mixing - "Fv" to be calculated

An example:

If values at design conditions are:

$T_1$ = Boiler Supply = 180°F

$T_2$ = System Supply = 140°F

$\Delta T_S$ = System $\Delta$T = 20°F

$F_1$ = System Flow = 30 GPM

$T_R$ = System Return = $T_2$ - $\Delta T_S$ = 120°F

To find the injection pump flow rate ($F_V$):

$$F_V = \frac{F_1 \times \Delta T_S}{T_1 - T_R}$$

$$F_V = \frac{30 \times 20}{180 - 120}$$

$$F_V = \frac{600}{60}$$

$$F_V = 10 \text{ GPM}$$

This means to properly operate this system, the variable speed injection circulator need only inject 10 GPM, a rate easily handled by most small circulators.

## System Set-Up

To set up a variable speed mixing system properly:

- Start the system with the balancing valve in the fully open position.
- With the system under full load - all zones calling for heat - close the valve until the variable speed circulator is running at about 90% (check readout on Wirsbo DuoMix 201 for circulator speed).

Figures 4 & 5 show the two most common piping layouts for Variable Speed Injection Mixing. Pay particular attention to the drop lines - or thermal traps - shown in the injection legs. These are particularly important to prevent "ghost flow" from the Primary loop into the Secondary loop. In some installations, flow control valves may also be necessary. Consult the circulator manufacturers' pump curve charts in order to determine the best circulator for the job. For specific applications of the Wirsbo DuoMix 201, refer to Chapter 10, pages 130-131. For specific boiler piping arrangements, refer to Chapter 11, pages 166-169. For wiring specifics, refer to Chapter 13, pages 202-205.

In the piping arrangement shown, the Variable Speed Injection circulators are plumbed in such a way as to limit head pressure in the injection legs to only a few feet at most. Use standard pressure drop calculations and equivalent length of feet charts for exact calculations if required.

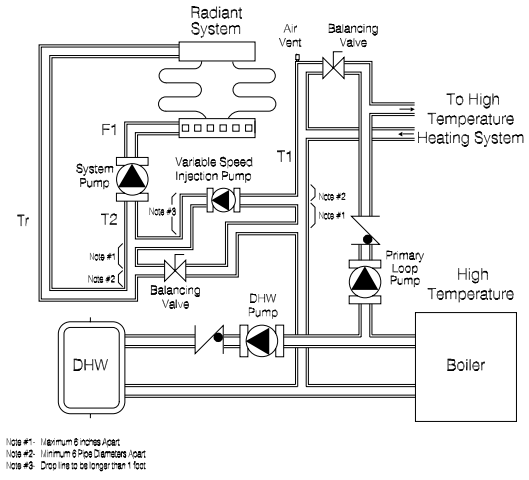



Figure 7 A typical system using Variable spped pump mixing through direct injection with high temp zones and DHW.



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# Variable Speed Injection Design Flow Rates

| Direct Injection Flow Rate (US GPM) | | Globe Valve Open Position (%) | Pipe Size | Injection Pump |
|---|---|---|---|---|
| No Glove Valve | With Globe Valve | | | |
| N/A | 1.5 | 20 | ½" | Grundfos 15-42 — Speed 2 |
| 4 | 3 | 100 | ½" | Grundfos 15-42 — Speed 2 |
| 4.5 | 3 | 50 | ½" | Taco 007 |
| 7.3 | 6 | 100 | ¾" | Grundfos 15-42 —Speed 2 |
| 9.5 | 7.4 | 100 | ¾" | Taco 007 |
| 10 | 8.3 | 100 | ¾" | Grundfos Brute or 15-42 — Speed 2 |
| 14 | 12 | 100 | 1" | Taco 007, Grundfos Brute or 15-42 — Speed 3” |
| 23 | 20 | 100 | 1¼" | Grundfos 26-64, Taco 0010” |
| 35 | 31 | 100 | 1½" | Grundfos 43-75, Taco 0012” |
| 41 | 38.5 | 100 | 2" | Grundfos 43-75F |
| 45 | 41.5 | 100 | 2" | Taco 0012 |

Table Courtesy Of tekmar
Table assumes 5 feet of pipe, 4 elbows and 4 branch tees of the listed diameter.
These circulators have been tested and approved by the manufacturers for use with tekmar variable speed electronics.

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